

Full length article

Adapting to drift: Weather-pattern experts for short-term photovoltaic power forecasting

Wentao Wang^{a,*,}, Haiyan Lu^a, Ayesha Ubaid^a, Fanyi Yang^b, Runyao Yu^{c,d},
Jianzhou Wang^e

^a School of Computer Science, University of Technology Sydney, Australia

^b School of Computer Science and Technology, Xidian University, China

^c Department of Electrical Sustainable Energy, Delft University of Technology, The Netherlands

^d Integrated Energy Systems, Austrian Institute of Technology, Austria

^e Department of Engineering Science, Macau University of Science and Technology, China

ARTICLE INFO

Keywords:

Photovoltaic power forecasting
Internal and external drift
Prototype-guided weather experts
Proactive update strategy

ABSTRACT

Photovoltaic (PV) power forecasting is often developed under an implicit assumption of stationarity, yet the weather–power relationship evolves over time and induces distribution shifts commonly known as concept drift. Existing PV power forecasting models either ignore this issue entirely or address it in a limited and insufficient manner. To better understand this non-stationarity, we distinguish between internal and external drift, which motivate the key components of our design. Therefore, we present **ADrift**, a drift-aware forecasting framework that integrates patch-based temporal modeling, weather-guided representation learning, and lightweight online adaptation. To model internal drift, the backbone employs prototype-guided weather experts that capture diverse meteorological patterns within each input window. To cope with external drift, a learnable adapter updates its parameters through a temporal gap attention mechanism that enables targeted adjustments to the model. In addition, a proactive update strategy further mitigates supervision delays under rapidly changing conditions. Experiments on three real-world PV datasets show that ADrift consistently improves forecasting accuracy over static and online-learning baselines, demonstrating its potential for practical deployment under evolving weather conditions.

1. Introduction

With the increasing availability of high-resolution PV monitoring data, short-term photovoltaic power forecasting has become a crucial task in modern power systems. However, PV power generation exhibits strong temporal variability driven by rapidly changing meteorological conditions, such as intermittent cloud cover and localized atmospheric dynamics [1]. These factors result in highly volatile power outputs, posing persistent challenges to forecasting accuracy and grid reliability, especially at fine temporal scales (e.g., 5-minute resolution) [2].

Recent research has increasingly focused on enhancing forecasting models, including recurrent neural networks (RNNs), temporal convolutional networks (TCNs), Transformers, and even large language models (LLMs). While these models have achieved notable accuracy gains, they largely overlook a fundamental challenge in PV power forecasting, that is, **concept drift**. In this context, concept drift refers to the temporal evolution of the conditional mapping between inputs and

outputs, such that $P(\mathcal{Y}|\mathcal{X})$ changes over time. The input $\mathcal{X}$ includes both meteorological features (e.g., solar radiation, temperature, wind speed, humidity, and so on) and historical PV power data, while the output $\mathcal{Y}$ represents future PV power. Concept drift implies that this conditional relationship is inherently non-stationary, as even near-identical meteorological inputs and historical PV levels can yield sharply divergent future trends. This phenomenon is primarily driven by the irregular movement of clouds and other transient atmospheric dynamics that rapidly alter the solar irradiance received by PV systems.

To clarify this point, our statistical analysis provides empirical evidence of such concept drift. Using Kernel Density Estimation (KDE) and Maximum Mean Discrepancy (MMD), we observe substantial discrepancies between hourly patches within the same day (Fig. 1). Elevated MMD values and structured off-diagonal blocks reveal that PV power generation within a single day originates from multiple, divergent weather patterns. Consequently, the mapping between meteorological

* Corresponding author.

E-mail addresses: wentao.wang@student.uts.edu.au (W. Wang), haiyan.lu@uts.edu.au (H. Lu), ayesha.ubaid@uts.edu.au (A. Ubaid), yangfanyi@stu.xidian.edu.cn (F. Yang), runyao.yu@tudelft.nl (R. Yu), jzwang@must.edu.mo (J. Wang).

<https://doi.org/10.1016/j.aei.2026.104781>

Received 13 January 2026; Received in revised form 29 March 2026; Accepted 4 May 2026

Available online 11 May 2026

1474-0346/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

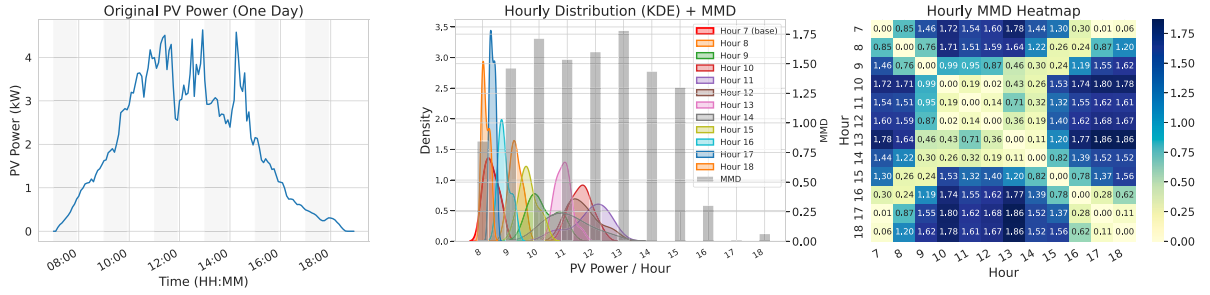


Fig. 1. Illustration of intra-day weather-pattern shifts in PV power data. **Left:** Five-minute PV power time series for a representative day, with hourly patches overlaid. Frequent instantaneous fluctuations, short drops, and recoveries highlight the inherent variability of PV power generation even within short intervals. **Middle:** Hourly KDE visualize the distinct distributions of these patches, while the gray bars indicate the MMD relative to the base hour. Several patches show large MMD values, confirming substantial intra-day weather-pattern shifts. **Right:** The MMD heatmap further reveals that, although adjacent patches often share similar distributions (light near-diagonal regions), many pairs of temporally distant patches diverge strongly (dark off-diagonal blocks).

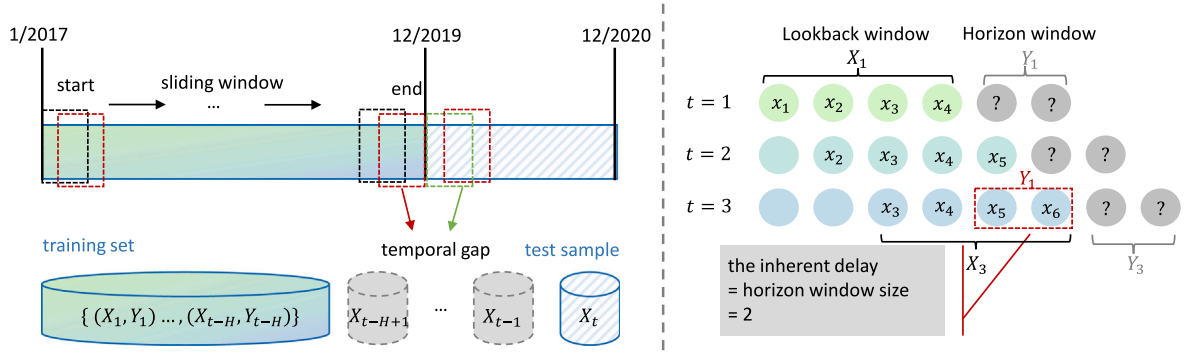


Fig. 2. Schematic illustration of the temporal gap (left) and inherent delay (right) arising from the daily sliding-window paradigm. **Left:** Because training and test sets are partitioned along the time axis under the sliding-window mechanism, the model is inherently trained on past distributions and evaluated on future ones, resulting in an unavoidable temporal gap. **Right:** For each training pair (X_t, Y_t) , the ground truth Y_t becomes available only after h time steps. For example, when the model receives (X_1, Y_1) , the corresponding label Y_1 appears in X_3 , thereby inducing an inherent delay between input and supervision.

variables and PV output is not stationary even within a diurnal cycle. We therefore instantiate this concept drift as a weather-pattern shift, a phenomenon where transitions between meteorological patterns (e.g., clear, cloudy, or overcast conditions) continuously reshape the underlying forecasting relationship.

Based on temporal scale, we further distinguish two manifestations of concept drift. The **internal drift** refers to the intra-day variability within the training data, where different time segments correspond to distinct meteorological patterns driven by evolving weather patterns. The **external drift**, in contrast, is used in a deployment-oriented sense to describe the mismatch induced by train-test temporal gaps and delayed supervision during online forecasting, rather than being restricted to the classical definition of concept drift as a shift in the underlying data-generating process. As illustrated in Fig. 2, the sliding-window mechanism inherently trains on past data and evaluates on future data, leading to an unavoidable temporal gap. Furthermore, the ground-truth labels for future samples become available only after the forecast horizon, introducing an inherent supervision delay. These factors do not directly cause new weather patterns but hinder the model's ability to adapt to newly emerging weather-pattern shifts, thereby limiting forecasting accuracy in real-world scenarios.

Taken together, short-term PV power forecasting is fundamentally challenged by a two-fold concept drift, comprising **internal drift** arising from evolving intra-day meteorological patterns and **external drift** occurring across training and testing periods, where temporal gaps and supervision delays hinder adaptation to new weather patterns. Along this line, a forecasting paradigm focused solely on static test-set accuracy, while neglecting these dual and interacting sources of concept drift, is fundamentally misaligned with the dynamic nature of real-world PV systems.

Therefore, this work aims to adapt forecasting models to concept drift by addressing two fundamental research questions: (1) How can we model the diverse intra-day patterns induced by dynamic meteorological conditions, rather than relying on a single stationary assumption? (2) How can the model adapt to newly arriving PV power data to mitigate the impact of temporal gaps and supervision delays? To answer these questions, we propose **ADrift**, short for **Adapting to Drift**, a weather-pattern expert model tailored for short-term PV power forecasting. For the **internal drift**, ADrift introduces weather prototype-guided experts equipped with a dynamic routing mechanism, enabling specialized experts to capture diverse and fine-grained meteorological patterns across different periods of the day. For the **external drift**, ADrift integrates a learnable adapter module that dynamically modulates the backbone's forward computation to align with newly emerging weather patterns. By leveraging both partially and fully observed ground truth, the adapter proactively updates its own parameters under supervision delays and generates modulation coefficients to construct effective weights for prediction. Extensive experiments on three real-world PV datasets demonstrate that ADrift consistently outperforms state-of-the-art baselines, particularly under dynamic and non-stationary conditions. In summary, the main contributions of this work are as follows:

1. We characterize short-term PV power forecasting as a problem involving by two forms of concept drift, corresponding to internal and external weather-pattern shifts. The internal drift is caused by fine-grained intra-day variability, while the external drift arises from temporal gaps and supervision delays. This formulation provides a unified perspective on non-stationarity in PV power forecasting and motivates our adaptive framework design.

2. To address the internal drift, we propose weather prototype-guided experts with a dynamic routing mechanism, which treats the daily forecasting task as a collection of patch-level sub-problems and enables specialized experts to capture intra-day concept drift.
3. To mitigate the external drift, we propose a learnable adapter that dynamically adjusts model parameters according to newly observed weather patterns. By updating parameters under both partially and fully observed supervision, the model can proactively adapt to new meteorological patterns even under supervision delays.

The remainder of this paper is organized as follows: Section 2 introduces related work. Section 4 presents the problem formulation and model design. Section 5 describes the experimental settings and analyzes the results, and finally, Section 7 concludes this work.

2. Related work

2.1. Short-term PV power forecasting

In general, short-term PV power forecasting aims to predict the future PV power trend one day ahead [3]. Early short-term PV power forecasting methods primarily rely on physical models, which use mathematical equations to estimate PV power based on geographic and meteorological factors such as irradiance, temperature, and wind speed [4]. While these models achieve satisfactory accuracy under stable weather conditions, they struggle with rapid meteorological fluctuations and require significant domain expertise. In contrast, data-driven models focus on learning the relationship between historical inputs (such as PV power and weather data) and future output power [5–8]. These models depend solely on past data, eliminating the need to account for the physical mechanisms of the system. For example, Lin et al. [5] applied temporal convolutional neural networks (TCN) to solar power forecasting. Agga et al. [6] later introduced two hybrid models, CNN-LSTM and ConvLSTM, to improve power forecasting accuracy for a self-consumption PV plant. More recently, Wang et al. [8] proposed a complementary time-frequency domain approach (CTFD) jointly models temporal and spectral variations to better capture PV power fluctuations and improve forecasting accuracy.

Despite their success, most existing approaches primarily focus on improving static forecasting accuracy while assuming that the underlying data distribution remains stationary. However, in real-world PV systems, the mapping from historical meteorological variables and past PV power to future PV generation is inherently non-stationary, continually influenced by evolving weather dynamics and seasonal variations. This phenomenon, known as *concept drift* [9], remains an underexplored issue in PV power forecasting. Nevertheless, a few recent studies have begun to explore this problem. Luo et al. [10] proposed an adaptive deep learning framework that retrains the model when prediction errors increase, while Zhang et al. [11] introduced an incremental learning method combining Broad Learning and Federated Learning for drift detection and online updating. Although these methods introduce adaptivity, they still treat concept drift as a generic statistical change rather than explicitly modeling how the weather–power relationship evolves. Moreover, their update mechanisms are reactive and coarse-grained, without distinguishing between the fine-grained intra-day fluctuations and the long-term cross-period shifts that jointly affect PV forecasting.

In contrast, our work explicitly interprets concept drift as a weather-pattern shift, which manifests in two complementary forms: (1) internal drift, driven by intra-day meteorological variability, and (2) external drift, caused by cross-period distributional gaps between training and deployment. To address these challenges, we employ a Mixture-of-Experts design to model internal variability and an online model adaptation mechanism to dynamically handle external drift. The following two subsections review related studies in these two directions, highlighting how existing methods have inspired our design yet remain limited in addressing concept drift in PV forecasting.

2.2. Mixture-of-experts

The Mixture-of-Experts (MoE) architecture was originally proposed to improve model scalability and specialization by decomposing complex mappings into multiple sub-functions, each handled by a specialized expert network [12,13]. Through conditional computation, a routing mechanism selectively activates a subset of experts, allowing the model to achieve high representational capacity while maintaining computational efficiency [14,15]. This conditional and modular design also provides flexibility for modeling heterogeneous or evolving patterns, making MoE particularly suitable for time-series forecasting under non-stationary conditions such as weather-driven PV power generation.

Building on this principle, recent studies have extended MoE structures to time-series forecasting to better capture non-stationarity and distributional heterogeneity. Sun et al. [16] proposed pattern-specific experts that explicitly model patch-level distribution shifts through a subspace-based Pattern Identifier, which clusters input patches and assigns each cluster to a dedicated expert. This design improves robustness to local distributional variations by enabling experts to specialize in distinct temporal segments. However, because the clustering process and expert assignments are explicitly defined, the model struggles to adapt smoothly when weather patterns evolve continuously within a day. To further enhance expert diversity and capture more complex multimodal behaviors, Wang et al. [17] introduced a dual-clustering MoE that jointly clusters temporal segments and feature channels. While this approach extends the idea of subspace-based specialization, it still relies on static cluster boundaries fixed after training, making it difficult to accommodate gradual distributional changes. Moreover, the learned clusters are purely data-driven, which limits their applicability to PV power forecasting where the weather–power relationship evolves dynamically over time.

The above limitations indicate that although MoE-based architectures are effective for modeling heterogeneous temporal behaviors, existing routing mechanisms remain discrete and static. To address this issue, we design a prototype-enhanced MoE that learns a compact set of prototypes serving as latent anchors for expert routing. Unlike clustering-based methods, the prototype module performs soft similarity-based routing, enabling smooth expert transitions as weather patterns change during the day. This mechanism allows each expert to implicitly specialize in a specific weather pattern while maintaining continuous adaptability, thereby effectively capturing intra-day concept drift in PV power forecasting. However, even prototype-enhanced MoE frameworks mainly focus on modeling internal variability and assume fixed expert configurations once trained. To handle external drift that evolves across longer periods, a complementary online adaptation mechanism is required, an aspect further discussed in the following section.

2.3. Online model adaptation

Online model adaptation [18] has increasingly become an effective strategy for handling evolving data distributions in time series forecasting. Recent methods such as FSNet [19] focus on updating the fast-evolving component of the model to enhance responsiveness to recent changes. This line of work is further extended by OneNet [20], which replaces component-wise updating with a more flexible expert-weighting mechanism that adjusts model inductive biases based on online forecasting feedback. Building upon the need to react more selectively to distributional changes, SOLID [21] detects latent contextual discrepancies and performs localized fine-tuning accordingly.

As online forecasting moved toward more practical streaming scenarios, later studies began to address the fundamental limitation of unavailable future supervision. Act-Now [22] and DSOF [23] both propose to generate pseudo labels for unobserved values and combine

them with observed labels to construct synthetic supervision for on-line updates. These designs help maintain chronological correctness and reduce information leakage, but their adaptation effectiveness is inevitably tied to the quality of pseudo labels and therefore remains sensitive to substantial concept drift.

More recent work shifts from supervision reconstruction toward directly modeling distribution evolution. TAFAS [24] adapts models at test time by updating a small calibration module using prediction-consistency signals computed on test samples, while PROCEED [25] explicitly estimates drift between consecutive samples and uses this estimate to proactively modulate model parameters. These approaches reduce reliance on labeled updates and provide more direct mechanisms for handling non-stationarity. However, the irregular and non-periodic nature of PV power generation, driven by rapid cloud-induced intra-day variability and slower seasonal shifts, challenges assumptions of stable contexts, smooth drift evolution or reliable pseudo label supervision. In this work, we instead perform adaptation at the patch level and restrict updates to a compact and parameter-efficient adapter guided by supervision delay signals, thereby offering an adaptation mechanism that is well suited to the drift characteristics of PV power forecasting.

3. Preliminary

We consider a short-term PV power forecasting task based on a multivariate time series $S \in \mathbb{R}^{N \times T}$, where N denotes the number of variables (i.e., $N-1$ meteorological features and 1 PV power series) and T is the total number of time steps. The objective is to learn a mapping function $f_\theta : \mathbb{R}^{N \times L} \rightarrow \mathbb{R}^{1 \times H}$ that predicts the PV power sequence over a forecasting horizon of length H based on an input lookback window of length L containing both PV power and meteorological features.

Formally, by applying a sliding window with one-step stride to S , we obtain a collection of paired samples $\mathcal{P} = \{(\mathcal{X}_t, \mathcal{Y}_t)\}_{t=1}^M$, where $\mathcal{X}_t \in \mathbb{R}^{N \times L}$ represents the lookback window and $\mathcal{Y}_t \in \mathbb{R}^{1 \times H}$ is the corresponding forecasting target. The standard learning objective is to minimize the empirical forecasting loss:

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{M} \sum_{t=1}^M \mathcal{L}(f_\theta(\mathcal{X}_t), \mathcal{Y}_t). \quad (1)$$

Unlike stationary time series forecasting, PV power generation exhibits strong non-stationarity due to the dynamic and highly variable nature of solar radiation. The statistical relationships between meteorological conditions and PV power outputs evolve over time, leading to *concept drift*, a temporal change in the conditional distribution $P(\mathcal{Y}|\mathcal{X})$ that governs the mapping from inputs to outputs.

Definition 1 (Concept Drift). *Concept drift refers to the temporal evolution of the underlying predictive function or conditional distribution $P(\mathcal{Y}|\mathcal{X})$. Formally, for two time indices t_1, t_2 within the observation period, concept drift occurs when*

$$P_{t_1}(\mathcal{Y}|\mathcal{X}) \neq P_{t_2}(\mathcal{Y}|\mathcal{X}). \quad (2)$$

In short-term PV power forecasting, such drift reflects changes in the physical and meteorological mechanisms driving solar power generation.

For analytical clarity, we distinguish two major forms of concept drift in PV power forecasting:

Definition 2 (Internal Drift). *In this work, internal drift refers to the within-day variation of the conditional mapping $P(\mathcal{Y}|\mathcal{X})$ induced by evolving weather conditions, rather than the deterministic intra-day periodicity itself. Even within the same day, rapid changes in solar irradiance, cloud motion, and atmospheric conditions may alter the relationship between meteorological inputs and PV output. Formally, let $\tau_1, \tau_2 \in \{1, \dots, L+H\}$ be two*

intra-day intervals of the same day d . Then, internal drift is characterized as

$$P_{d, \tau_1}(\mathcal{Y}|\mathcal{X}) \neq P_{d, \tau_2}(\mathcal{Y}|\mathcal{X}), \quad (3)$$

where the change is attributed to weather-driven mapping variation beyond regular daily seasonality.

Definition 3 (External Drift). *External drift refers to the inter-day evolution of the conditional mapping $P(\mathcal{Y}|\mathcal{X})$ between the training and deployment periods. It reflects the gradual change of the global feature-target relationship across days, which leads to train-test mismatch when a model trained on past data is applied to future data. Let d_{tr} and d_{te} denote the training and testing periods ($d_{tr} \neq d_{te}$). Then*

$$P_{d_{tr}}(\mathcal{Y}|\mathcal{X}) \neq P_{d_{te}}(\mathcal{Y}|\mathcal{X}). \quad (4)$$

In practical forecasting pipelines, supervision delays further increase the difficulty of adapting the model to such inter-day changes in a timely manner.

Under such non-stationary conditions, the forecasting objective can be reformulated as

$$\hat{\theta} = \arg \min_{\theta} \mathbb{E}_{(\mathcal{X}_t, \mathcal{Y}_t) \sim P_t(\mathcal{X}, \mathcal{Y})} [\mathcal{L}(f_\theta(\mathcal{X}_t), \mathcal{Y}_t)], \quad (5)$$

where $P_t(\mathcal{X}, \mathcal{Y})$ denotes the time-dependent joint distribution at time t . This dynamic setting highlights the fundamental challenge of PV power forecasting: the model are supposed to continuously adapt to evolving concepts, including both the fine-grained intra-day drift and the long-term cross-period drift, thereby motivating the development of a dual-drift-aware forecasting framework.

4. Methodology

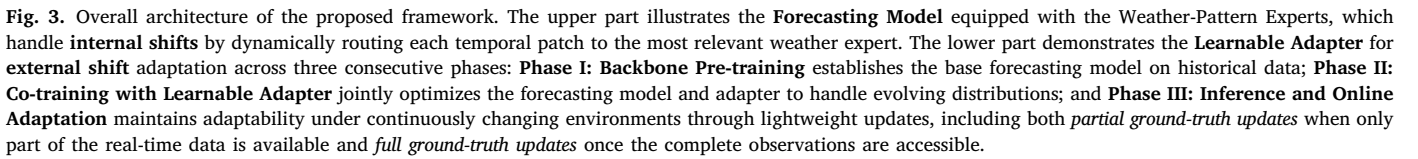
4.1. Overview

This section provides an overview of our proposed framework for short-term PV power forecasting under real-world non-stationary conditions. Taken together, the core challenge in this task lies in a two-fold concept drift that existing paradigms largely overlook: an internal drift driven by evolving intra-day distributions (e.g., rapid irradiance fluctuations and transient cloud movements), and an external drift induced by temporal train-test gaps and supervision delays that hinder consistent model adaptation. To handle the two-fold concept drift, our framework adopts a modular architecture that integrates two complementary components. First, the weather-pattern expert module (Section 4.2) captures intra-day variations by assigning each temporal patch to its most representative weather expert through a prototype-based top- k routing mechanism, thereby mitigating the internal drift. Second, the learnable adapter module (Section 4.3) addresses the external drift caused by temporal gaps and delayed supervision, by introducing a layer-wise bottleneck generator and a temporal gap attention mechanism that enable lightweight, adaptive update without full retraining.

During training and inference (Section 4.4), these two modules operate jointly in an end-to-end manner: the prototype-guided weather experts learn to represent short-term intra-day dynamics, while the learnable adapter continuously aligns the forecasting model with evolving data distributions caused by external drift. An overview of the complete architecture is illustrated in Fig. 3.

4.2. Weather-pattern experts for concept internal drift

To address the internal concept drift in PV power forecasting, we introduce a weather-pattern expert module that dynamically assigns each patch to the most suitable expert, enabling the model to capture fine-grained, intra-day weather-induced variations.



5

where $\mathbf{W}_1^{(e)} \in \mathbb{R}^{d \times d}$ and $\mathbf{W}_2^{(e)} \in \mathbb{R}^{d \times d}$. Each expert produces an output $\mathbf{h}_i^{(e)}$ with the same shape $\mathbb{R}^{L_p \times d}$ as the input. Next, we adopt a top- k sparse routing strategy, where only the k experts with the highest logits are activated for each patch. Formally, let $\mathcal{K}_i = \arg \text{top-}k(\mathbf{u}_i)$ denote the index set of the top- k experts. The sparse expert weights are computed as:

$$w_{i,e} = \begin{cases} \frac{\exp(u_{i,e})}{\sum_{e' \in \mathcal{K}_i} \exp(u_{i,e'})}, & e \in \mathcal{K}_i, \\ 0, & \text{otherwise,} \end{cases} \quad (12)$$

and the resulting sparse weight vector is denoted as $\mathbf{w}_i \in \mathbb{R}^E$. This routing mechanism maintains the flexibility of a mixture-of-experts structure while reducing the computational cost from $\mathcal{O}(EL_p d^2)$ to $\mathcal{O}(kL_p d^2)$. In this way, the final patch-level representation is obtained by aggregating only the top- k activated experts:

$$\mathbf{o}_i = \sum_{e \in \mathcal{K}_i} w_{i,e} \mathbf{h}_i^{(e)}. \quad (13)$$

Since $\mathbf{o}_i \in \mathbb{R}^{L_p \times d}$, the dimensional consistency with $\hat{\mathbf{X}}_i$ is preserved, enabling smooth concatenation in the subsequent forecasting stage.

4.2.4. Forecasting

After expert aggregation, we obtain a set of patch-level outputs $\{\mathbf{o}_i\}_{i=1}^{N_p}$, where each $\mathbf{o}_i \in \mathbb{R}^{L_p \times d}$. Since all patches share the same latent embedding dimension d and are non-overlapping, the full temporal representation is obtained by concatenating the patch outputs along the temporal axis $\mathbf{O} = \text{Concat}([\mathbf{o}_1, \mathbf{o}_2, \dots, \mathbf{o}_{N_p}])$. The resulting $\mathbf{O}$ has the shape $\mathbb{R}^{L \times d}$, corresponding to the complete latent temporal representation of the lookback window. Finally, a lightweight nonlinear prediction head $g(\cdot)$ maps $\mathbf{O}$ to the forecasted PV power sequence:

$$\hat{\mathbf{Y}} = g(\mathbf{O}) = \text{ReLU}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{O} + \mathbf{b}_1) + \mathbf{b}_2), \quad (14)$$

where $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times 1}$ denotes the predicted PV power sequence over the forecasting horizon H . This expert aggregation allows the model to respond adaptively to intra-day weather pattern shifts, thereby mitigating internal concept drift in the forecasting process.

4.3. Learnable adapter for external concept drift

While the *internal concept drift* is considered within weather-pattern experts, the forecasting backbone may still suffer from performance degradation when facing *external concept drift*, which, in our single-site setting, arises from inter-day train-test gaps and supervision delays during the test stage. Such drift alters the global input-output mapping, and the parameters learned from the training stage may no longer generalize to unseen weather patterns. To enhance adaptability without retraining the entire model, we introduce a learnable adapter that dynamically reparameterizes key linear layers of the backbone. The adapter acts as an external plug-in module that interprets the temporal relationship between consecutive samples and generates lightweight, layer-wise adjustments aligned with the current data structure.

4.3.1. Temporal gap attention

To cope with the temporal gap and delayed supervision during the test stage, we design a Temporal Gap Attention (TGA) mechanism to extract a drift signal with a fixed lag H as illustrated in Fig. 2. Unlike direct element-wise differences that enforce one-to-one correspondence across positions, TGA performs content-based soft matching between the *current* sample and a *lagged* sample, thereby providing a more reasonable estimate of structural changes. Specifically, let $\hat{\mathbf{X}}_t \in \mathbb{R}^{N_p \times L_p \times d}$ denote the patch-level embeddings of the lookback window ending at time t , and let $\hat{\mathbf{X}}_{t-H} \in \mathbb{R}^{N_p \times L_p \times d}$ denote the embeddings of the lookback window at time $t-H$, both obtained from the same temporal encoder (Section 4.2.1). We first project both embeddings into a reduced latent space ($d_r \ll d$):

$$\mathbf{Q} = \hat{\mathbf{X}}_t \mathbf{W}_Q, \quad \mathbf{K} = \hat{\mathbf{X}}_{t-H} \mathbf{W}_K, \quad \mathbf{V} = \hat{\mathbf{X}}_{t-H} \mathbf{W}_V, \quad (15)$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times d_r}$. When computing attention, we flatten the patch and within-patch dimensions so that $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{(N_p L_p) \times d_r}$, and reshape the output back afterward. The aligned historical representation is computed as:

$$\mathbf{H}_{att} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_r}}\right)\mathbf{V}, \quad (16)$$

producing $\mathbf{H}_{att} \in \mathbb{R}^{N_p \times L_p \times d_r}$. Finally, we calculate the drift representation as the residual between the current embedding and the attended lagged reference:

$$\mathbf{E}_{gap} = \mathbf{Q} - \mathbf{H}_{att} \in \mathbb{R}^{N_p \times L_p \times d_r}. \quad (17)$$

By global average aggregation over all patch and within-patch positions, we obtain a compact drift vector $\bar{\mathbf{E}}_{gap} = \text{Average}(\mathbf{E}_{gap}) \in \mathbb{R}^{d_r \times 1}$, which is then fed into the bottleneck generator to adjust the adapter parameters.

4.3.2. Layer-wise bottleneck parameter generator

While the TGA module captures the contextual discrepancy between two lookback windows separated by a fixed lag H , the model still requires a mechanism to translate this observed drift into concrete parameter adjustments within the forecasting backbone [25]. To achieve this, we adopt a layer-wise bottleneck generator that maps the drift embedding into adaptive modulation coefficients for each linear layer.

Given the drift representation $\bar{\mathbf{E}}_{gap}$ produced by TGA, which summarizes the feature-wise discrepancy between the current window ending at t and the lagged reference window at $t-H$ (i.e., between $\hat{\mathbf{X}}_t$ and $\hat{\mathbf{X}}_{t-H}$), the generator produces two sets of coefficients, $\alpha_t^{(l)} \in \mathbb{R}^{d_{in} \times 1}$ and $\beta_t^{(l)} \in \mathbb{R}^{d_{out} \times 1}$, for each linear projection layer l (i.e., `nn.Linear`) in the backbone. These coefficients modulate the base parameters to better match the data distribution observed at time t . Formally, let $\theta_{base}^{(l)} \in \mathbb{R}^{d_{in} \times d_{out}}$ denote the base parameters of the backbone's l th linear layer (obtained from offline training or the latest full-update checkpoint during test stage). The adaptation generator computes:

$$[\alpha_t^{(l)}, \beta_t^{(l)}] = (\mathbf{W}_2^{(l)})^\top \sigma\left(\mathbf{W}_1^{(l)} \bar{\mathbf{E}}_{gap} + \mathbf{b}_1^{(l)}\right) + \mathbf{b}_2^{(l)}, \quad (18)$$

where $\mathbf{W}_1^{(l)} \in \mathbb{R}^{r \times d_r}$ and $\mathbf{W}_2^{(l)} \in \mathbb{R}^{(d_{in}+d_{out}) \times r}$ form a bottleneck structure with a small hyperparameter $r \ll \min(d_{in}, d_{out})$, $\sigma(\cdot)$ is a sigmoid activation, and $\mathbf{b}_1^{(l)}, \mathbf{b}_2^{(l)}$ are learnable biases. To maintain parameter efficiency, we employ shared projection matrices $\mathbf{W}_1^{(l)}$ and $\mathbf{W}_2^{(l)}$ among layers of the same type (e.g., projection layers), while individual biases provide layer-specific flexibility.

The effective parameters for the l th layer at time t are obtained by scaling the base weights with the two modulation vectors:

$$\theta_t^{(l)} = (\alpha_t^{(l)} (\beta_t^{(l)})^\top) \odot \theta_{base}^{(l)}, \quad (19)$$

where $\odot$ denotes element-wise multiplication. Through this mechanism, the adaptation generator generates modulation coefficients for time t conditioned on the lag- H drift embedding $\bar{\mathbf{E}}_{gap}$. Here, θ_{base} refers to the backbone parameters used as the modulation base. During offline training, it denotes the parameters optimized via mini-batch gradient descent. During online inference, θ_{base} remains fixed and serves as a stable modulation base, and it is not updated by gradient descent.

Because the projection matrices $\mathbf{W}_1^{(l)}$ and $\mathbf{W}_2^{(l)}$ are shared across layers of the same type, the number of weight parameters does not scale with the number of adapted layers N_ℓ . The total parameter count is determined by the shared weights and the N_ℓ individual bias terms:

$$\underbrace{\mathcal{O}(r(d_r + d_{in} + d_{out}))}_{\text{Shared Weights } (\mathbf{W}_1, \mathbf{W}_2)} + \underbrace{\mathcal{O}(N_\ell(r + d_{in} + d_{out}))}_{\text{Individual Biases } (\mathbf{b}_1, \mathbf{b}_2)}.$$

This count is dominated by the individual bias terms, $\mathcal{O}(N_\ell(d_{in} + d_{out}))$, and is a significant reduction compared to the fully connected case $\mathcal{O}(N_\ell d_r d_{in} d_{out})$ or a case where all weights are individual. This shared-weight design reduces redundancy while preserving sufficient layer-specific flexibility through independent bias terms, keeping the adapter parameter-efficient for online deployment.

4.4. Mini-batch training and inference phase

This section introduces a three-phase training and inference strategy that jointly addresses both *internal* and *external* concept drift in PV power forecasting. Phase I pre-trains the backbone (weather-pattern experts) to capture stable intra-day patterns, thereby mitigating internal drift caused by short-term weather fluctuations. Phase II co-trains the learnable adapter with the backbone, enabling it to generate layer-wise modulation coefficients driven by a fixed-lag drift signal. Phase III performs online inference with delayed supervision: the backbone is kept stable, while the adapter is updated only when scheduled or triggered, combining periodic full updates (as commonly adopted in practice) with FFT-driven partial updates in-between. An overview of the overall procedure is summarized in Algorithm 1.

4.4.1. Phase I: Backbone pre-training

In the first phase, the backbone model f_{θ_b} is trained on the training set $\mathcal{D}_{train}$ while the adapter g_{θ_a} is frozen. The goal is to learn a stable mapping that captures internal weather variations within each day. The model minimizes the mean squared error (MSE) $\mathcal{L}(f_{\theta_b}(\mathcal{X}), \mathcal{Y})$ over mini-batches and updates θ_b by gradient descent. This phase produces a generalized forecasting backbone for subsequent co-training.

4.4.2. Phase II: Co-training with learnable adapter

Building on the parameter modulation mechanism in Section 4.3.2, this stage jointly optimizes the forecasting backbone and the learnable adapter. The adapter translates the drift captured by the TGA module into layer-wise modulation of the backbone parameters.

Given a mini-batch $\mathcal{B}_k = \{(\mathbf{X}_{k,j}, \mathbf{Y}_{k,j})\}_{j=1}^B$ sampled in chronological order, for each sample indexed by its window-ending time t we retrieve a lagged reference window ending at $t - H$ from a temporal buffer, and compute the drift embedding $\tilde{\mathbf{E}}_{gap}(t)$ between $\tilde{\mathbf{X}}_t$ and $\tilde{\mathbf{X}}_{t-H}$. For each backbone layer l , the bottleneck generator $G^{(l)}(\cdot)$ maps $\tilde{\mathbf{E}}_{gap}(t)$ to modulation coefficients $(\alpha_t^{(l)}, \beta_t^{(l)})$, which are applied to obtain the effective parameters $\theta_t^{(l)}$ used for forecasting at time t .

The forecasts are computed as $\hat{\mathbf{Y}}_{k,j} = f_{\theta_b, \theta_a}(\mathbf{X}_{k,j})$ and the batch loss $\mathcal{L}_k = \frac{1}{B} \sum_j \|\hat{\mathbf{Y}}_{k,j} - \mathbf{Y}_{k,j}\|_2^2$ is used to jointly update both modules (backbone parameters θ_b and learnable parameters θ_a) by gradient descent. This enables the adapter to learn how to modulate the backbone conditioned on lag- H drift. Due to the shared bottleneck projection, the added computation grows only linearly with the number of layers, maintaining efficiency for end-to-end training. After convergence, the backbone is fixed as a temporally general forecasting base, while the adapter remains trainable during inference for online alignment to new temporal patterns.

4.4.3. Phase III: Inference and online adaptation

During inference, the backbone parameters θ_b are frozen, while the adapter remains trainable to accommodate unseen external drift. At each time step t , we compute $\tilde{\mathbf{E}}_{gap}(t)$ from the current window ending at t and the lagged reference window ending at $t - H$, then the adapter outputs modulation coefficients (α_t, β_t) to construct the effective weights used in the current forward pass, yielding the forecast $\hat{\mathbf{Y}}_t = f_{\theta_b}(\mathcal{X}_t)$. Importantly, the coefficients are generated at every step, whereas gradient-based updates to the adapter are performed only when triggered.

Partial update (FFT-triggered). Due to delayed supervision, waiting for the complete horizon label can slow down adaptation under rapidly changing conditions. We therefore apply additional *partial* updates using an FFT-based trigger. For each incoming batch, we use the first input window to estimate the short-term variability of the current segment. After mean normalization, the PV power sequence is transformed by FFT to obtain its spectrum, and the dominant frequency f_{max} yields an update interval $t_0 = 1/f_{max}$. When t_0 steps have elapsed since the last partial update, we update the adapter using the currently available

supervision (e.g., the visible prefix of the horizon²), while using the generated coefficients to re-parameterize the forward computation of the backbone without altering the stored base weights.

Full update (label-arrival triggered). Since training pairs are constructed by a sliding-window mechanism, the complete horizon label for a past input window becomes available only after all H future steps have been observed. Once such delayed supervision arrives at time t , we query the buffer to identify whether it completes the horizon of a previously stored lookback window that has not yet received a full update. If a match is found, we immediately perform a *full update* of the adapter using the complete ground truth for that window, and mark it as “fully updated” in the buffer to avoid repeated updates. Both partial and full updates share the same optimization rule and learning rate; the difference lies in the completeness of supervision (prefix versus full horizon) and the triggering condition (FFT-based versus label-arrival). In both cases, only the adapter parameters are updated, while the backbone base weights remain fixed.

Adaptive re-parameterization. In both update modes, the adapter is first updated (when triggered) and then regenerates the coefficients (α, β) for subsequent steps. These coefficients re-parameterize the forward computation of the backbone by constructing effective weights, while keeping the stored base parameters unchanged. This design maintains a stable global representation while enabling continuous alignment to external drift under delayed supervision.

4.5. Computational complexity analysis

We analyze the computational complexity of the proposed framework. Let L denote the lookback length, L_p the patch length, $N_p = L/L_p$ the number of patches, d the embedding dimension, h the expert hidden size, K the number of prototypes, k the number of active experts, d_r the TGA reduced dimension, r the bottleneck dimension, and N_ℓ the number of adapted layers. The per-sample complexity is the sum of its components:

- **Patch-based Time Encoder:** Applies a Transformer block (self-attention and FFN) to N_p patches of length L_p . The self-attention cost is $\mathcal{O}(N_p \cdot L_p^2 d) = \mathcal{O}(LL_p d)$. The feed-forward network cost is $\mathcal{O}(N_p \cdot L_p d^2) = \mathcal{O}(Ld^2)$. Total cost is $\mathcal{O}(LL_p d + Ld^2)$.
- **Weather Prototype Learning:** Involves computing $N_p \times K$ patch-prototype similarities, each requiring summation over L_p time steps of d -dimensional dot products. The cost is $\mathcal{O}(N_p \cdot K \cdot L_p \cdot d) = \mathcal{O}(LKd)$.
- **Expert Routing:** Activates k experts (MLPs) for N_p patches. The cost per expert is $\mathcal{O}(L_p d^2)$. The total cost is $\mathcal{O}(N_p \cdot k \cdot L_p d^2) = \mathcal{O}(Lkd^2)$.
- **Learnable Adapter:** This module processes sequences of total length L . The projections cost $\mathcal{O}(Ldd_r)$. The attention matrix computation $\mathbf{QK}^T$ and the subsequent multiplication by $\mathbf{V}$ are the dominant bottlenecks, costing $\mathcal{O}(L^2 d_r)$. The total for TGA is $\mathcal{O}(Ldd_r + L^2 d_r)$.
- **Bottleneck Generator:** Adds a minor cost of $\mathcal{O}(N_\ell(d_r r + rd))$, which is negligible compared to the sequence-dependent terms.

Combining these terms, the overall per-sample complexity is:

$$\mathcal{O}((LL_p d + Ld^2) + LKd + Lkd^2 + (Ldd_r + L^2 d_r)).$$

Since L_p, K, k, h, d_r are typically small constants, the complexity is dominated by the TGA module and the feed-forward networks in the encoder. The final complexity simplifies to $\mathcal{O}(L^2 d_r + Ld^2)$.

² The visible prefix refers to the subset of horizon labels that have already arrived through delayed real-time feedback and are retrieved from the historical buffer, and it does not contain any unobserved future ground truth.

Algorithm 1: Three-phase Training and Inference Strategy of ADrift

Input: Training set $\mathcal{D}_{train}$, testing stream $\mathcal{D}_{test}$, Learning rate η
Output: Forecasting results $\hat{\mathbf{Y}}$

- 1 Initialize backbone parameters θ_b and adapter parameters θ_a ;
- 2 Define Loss function $\mathcal{L}(\hat{\mathbf{Y}}, \mathbf{Y}) = \text{MSELoss}(\hat{\mathbf{Y}}, \mathbf{Y})$;
- 3 **Phase I: Backbone Pre-training**
- 4 **for** each iteration k with batch $B_k = (\mathbf{X}_k, \mathbf{Y}_k) \in \mathcal{D}_{train}$ **do**
- 5 Freeze adapter θ_a ;
- 6 $\hat{\mathbf{Y}}_k = f_{\theta_b}(\mathbf{X}_k)$;
- 7 $\mathcal{L}_k^{(I)} = \mathcal{L}(\hat{\mathbf{Y}}_k, \mathbf{Y}_k)$;
- 8 $\theta_b.\text{zero_grad}()$;
- 9 $\mathcal{L}_k^{(I)}.\text{backward}()$;
- 10 $\theta_b \leftarrow \theta_b - \eta \nabla_{\theta_b} \mathcal{L}_k^{(I)}$;
- 11 **Phase II: Co-training with Learnable Adapter**
- 12 **for** each iteration k with batch $B_k = (\mathbf{X}_k, \mathbf{Y}_k) \in \mathcal{D}_{train}$ **do**
- 13 Retrieve lagged context batch $\mathbf{X}_{k,\text{lag}}$ from buffer;
- 14 Compute drift $\bar{\mathbf{E}}_{\text{gap}} = \text{TGA}(\hat{\mathbf{X}}_k, \hat{\mathbf{X}}_{k,\text{lag}})$;
- 15 Generate coefficients $(\alpha_k, \beta_k) = G(\bar{\mathbf{E}}_{\text{gap}})$;
- 16 $\bar{\theta}_k = (\alpha_k \beta_k^T) \odot \theta_b$;
- 17 $\hat{\mathbf{Y}}_k = f_{\bar{\theta}_k}(\mathbf{X}_k)$;
- 18 $\mathcal{L}_k^{(II)} = \mathcal{L}(\hat{\mathbf{Y}}_k, \mathbf{Y}_k)$;
- 19 $(\theta_b, \theta_a).\text{zero_grad}()$;
- 20 $\mathcal{L}_k^{(II)}.\text{backward}()$;
- 21 $\theta_b \leftarrow \theta_b - \eta \nabla_{\theta_b} \mathcal{L}_k^{(II)}$;
- 22 $\theta_a \leftarrow \theta_a - \eta \nabla_{\theta_a} \mathcal{L}_k^{(II)}$;
- 23 **Phase III: Inference and Online Adaptation**
- 24 **for** each time step t with input $\mathbf{X}_t \in \mathcal{D}_{test}$ **do**
- 25 Freeze θ_b and keep θ_a trainable;
- 26 Retrieve lagged context $\mathbf{X}_{t-H}$;
- 27 // 1. Forecasting (following the procedure in Lines 13--17)
- 28 $\hat{\mathbf{Y}}_t = f_{\bar{\theta}_t}(\mathbf{X}_t)$;
- 29 // 2. Trigger Online Updates
- 30 Estimate dominant frequency $f_{\max}$ from $\mathbf{X}_t$; set $t_0 = 1/f_{\max}$;
- 31 **if** elapsed time $\geq t_0$ (Partial Update) **then**
- 32 Select visible prefix $h' = \min(\lfloor t_0 \rfloor, H)$;
- 33 $\mathcal{L}_t^{(III)} = \mathcal{L}(\hat{\mathbf{Y}}_{t[:h]}, \mathbf{Y}_{t[:h]})$;
- 34 $\theta_a.\text{zero_grad}()$;
- 35 $\mathcal{L}_t^{(III)}.\text{backward}()$;
- 36 $\theta_a \leftarrow \theta_a - \eta \nabla_{\theta_a} \mathcal{L}_t^{(III)}$;
- 37 **if** past sample $\mathbf{X}_{t'}$ has complete $\mathbf{Y}_{t'}$ (Full Update) **then**
- 38 Re-compute modulation $\bar{\theta}_{t'}$ for past sample $\mathbf{X}_{t'}$;
- 39 $\mathcal{L}_{t'}^{(III)} = \mathcal{L}(f_{\bar{\theta}_{t'}}(\mathbf{X}_{t'}), \mathbf{Y}_{t'})$;
- 40 $\theta_a.\text{zero_grad}()$;
- 41 $\mathcal{L}_{t'}^{(III)}.\text{backward}()$;
- 42 $\theta_a \leftarrow \theta_a - \eta \nabla_{\theta_a} \mathcal{L}_{t'}^{(III)}$;

5. Experiment

In this section, we conduct extensive experiments to evaluate the effectiveness of the proposed **ADrift** framework. Guided by the two research questions introduced earlier (see Section 1), our experimental analysis is first designed to address the following sub-questions:

- **Q1:** How does the static backbone of **ADrift** compare with existing static forecasting baselines (corresponding to RQ1)?

- **Q2:** How does the full **ADrift** framework compare with static baselines augmented by online learning mechanisms (corresponding to RQ2)?

We then take a further step to explore additional aspects of the proposed framework, aiming to provide further insights into its design:

- **Q3:** How do different components contribute to the overall performance of **ADrift**?
- **Q4:** How sensitive is **ADrift** to key hyperparameters?

5.1. Experimental setup

5.1.1. Dataset descriptions

In our experiments, we employ three PV power datasets from geographically diverse regions across both the Southern and Northern Hemispheres, including Alice Springs, Uluru, and PVOD. This geographical diversity helps to evaluate the generalizability of the proposed framework under different climate conditions. Their details are presented below.

For the first dataset, we use data from the '8' PV system at the DKASC site in Alice Springs,³ chosen for its relatively complete records with minimal faults. Following [26], we analyze data from March 1, 2019, to March 31, 2020, at a 5-minute resolution. Situated in the desert region of Alice Springs, this site is among the most solar-rich areas on Earth. Given that PV power output is typically low or near zero in the early morning and late evening, we focus on data collected between 7 AM and 7 PM, yielding a filtered dataset of 54,961 instances. For preprocessing, missing values are handled using forward filling. Additionally, following [27], we compute the Spearman and Pearson correlation coefficients between PV power and meteorological variables. As shown in Fig. 4(a), relative humidity, daily rainfall, and wind direction exhibit negative correlations with PV power, indicating that PV power decreases as these factors increase. In contrast, temperature, global horizontal radiation, and diffuse horizontal radiation show positive correlations, suggesting an increase in PV power with higher values of these factors. Considering both correlation analyses, we select global horizontal radiation and diffuse horizontal radiation ($|\rho| \geq 0.5$) as the primary influencing factors in the Alice Springs dataset.

The second dataset is Uluru,⁴ consisting of three years of data recorded from Site-3 A of the Uluru solar system located near Uluru, Australia. The meteorological data are from January 1, 2017 to December 31, 2020, with a resolution of 1 h. The data from the PV system are located in the desert plain with abundant sunshine during the day. Similarly, we also focus on data collected between 7 AM and 7 PM, yielding a filtered dataset of 17,485 sample pairs. In Fig. 4(b), the Spearman and Pearson correlation coefficients between PV power and meteorological variables, including Active Energy Delivered Received, Wind Speed, Weather Temperature, Global Horizontal Radiation, Wind Direction, Weather Daily Rainfall, Max Wind Speed, Air Pressure, Hail Accumulation, Pyranometer, Temperature Probe 1, Temperature Probe 2 and Active Power. Among them, Active Power represents the amount of electricity actually converted and transmitted by the PV system, and Pyranometer is the solar radiation sensor installed on the PV array, which is used to measure the real-time solar radiation received by the surface of the PV module.

The third public dataset, PVOD [28], consists of data from multiple PV power plants across different latitudes and longitudes in Hebei Province, northern China. In this study, we use data from the '1' PV station, as it contains no missing values. The dataset spans June 30, 2018, to June 12, 2019, with a 15-minute resolution. To ensure data relevance, we retain only observations between 5 AM and 8 PM daily, resulting in a filtered dataset of 20,761 instances. Additionally,

³ <https://dkasolarcentre.com.au/download?location=alice-springs>

⁴ <https://dkasolarcentre.com.au/download?location=yulara>

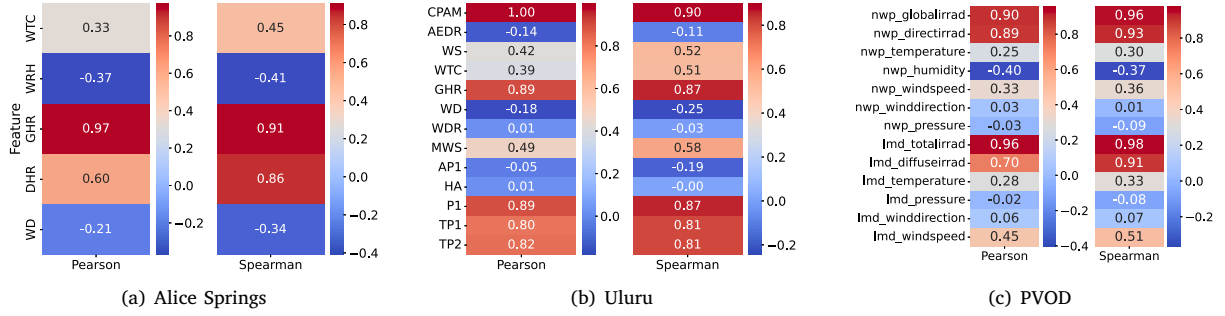


Fig. 4. Spearman and Pearson correlation coefficients between PV power and meteorological factors across three datasets. Factors with both correlation coefficients satisfying $|\rho| \geq 0.5$ are selected as inputs to the forecasting models.

this dataset includes both numerical weather prediction (NWP) data from meteorological services and local measurement data (LMD) from PV power stations. Similarly, we analyze the Spearman and Pearson correlation coefficients between PV power and LMD variables, as shown in Fig. 4(c). For the PVOD dataset, future NWP data are used as input because this dataset explicitly provides NWP information, which serves as an available forecasting prior.

Moreover, following [17,29], we apply MinMaxScaler to normalize three datasets and split them into training, validation, and test sets in a 7:2:1 ratio.⁵ To further assess our model's performance not only on the entire test set but also under different weather conditions, we follow the approach in [2] to identify 10 clear sky days and 10 cloudy sky days from the test sets of each dataset. This selection is based on historical weather data obtained from a free weather API,⁶ which provides daily climate classifications for any given location.

5.1.2. Implementation details

All experiments were conducted using PyTorch 2.4.0 and Python 3.10.14. Data preprocessing, including normalization and temporal slicing, was performed on an AMD EPYC 9354P (32-core, 3.25 GHz) CPU, while model training and inference were executed on an NVIDIA L4 GPU with 24 GB of memory. We adopted the Adam optimizer and applied early stopping based on the validation loss. To mitigate overfitting, dropout and weight decay regularization were applied during training. Each experiment was repeated five times with different random seeds, and the average performance was reported. All hyperparameters were tuned on the validation set within the search space summarized in Table 1, and the final configurations adopted for each dataset are reported below.

Optimal configuration. A comprehensive grid search was conducted across all datasets and training stages to determine the optimal model hyperparameters within the search space summarized in Table 1. While the overall architecture and optimization protocol remained consistent, the final configuration for each dataset was slightly adjusted to reflect its temporal resolution and meteorological variability. For Uluru, a higher-capacity setting with $d = 64$, $h = 2$, $K = 6$, $k = 2$, $E = 4$, $d_r = 24$, and $r = 12$ was found to improve stability. For PVOD, a diversified expert configuration ($d = 96$, $h = 2$, $K = 5$, $k = 2$, $E = 6$, $d_r = 24$, and $r = 16$) was empirically more effective in handling highly variable cloudy conditions. For Alice Springs, the optimal configuration adopted $d = 128$, $h = 2$, $K = 6$, $k = 2$, $E = 4$, $d_r = 48$, and $r = 24$. Across

Table 1

Hyperparameter search space used in all experiments. A grid search strategy was applied across all stages, including both pre-training and online adaptation.

Category	Search space
Optimization	
Optimizer	{Adam, AdamW}
Learning rate	$\{1 \times 10^{-5}, 3 \times 10^{-5}, 1 \times 10^{-4}, 3 \times 10^{-4}, 5 \times 10^{-4}\}$
Weight decay	$\{1 \times 10^{-6}, 1 \times 10^{-5}, 1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}\}$
Batch size	{16, 32, 64}
Global dropout	{0.1, 0.2, 0.3}
Windows	
Lookback L /Horizon H	{(144, 144), (12, 12), (60, 60)} for {Alice Springs, Uluru, PVOD}
Patch length L_p	{6, 12, 24}
Backbone (Transformer)	
Embedding dimension d	{64, 96, 128, 192}
Experts/Prototypes	
Prototypes K	{4, 5, 6, 7, 8}
Routing (top- k) & Experts E	$k \in \{1, 2\}, E \in \{2, 3, 4, 5, 6, 7\}$
Adapter (External Drift)	
Reduced dimension d_r	{16, 24, 32, 48, 64}
Bottleneck dimension r	{8, 12, 16, 24, 32}

all datasets, the lookback window L and forecasting horizon H were aligned to one day of data ($L = H = \{144, 12, 60\}$ for Alice Springs, Uluru, and PVOD, respectively). The models were optimized using AdamW with an initial learning rate of 5×10^{-4} , a weight decay of 1×10^{-3} , and a batch size of 32. A dropout rate of 0.3 was applied to the backbone and expert modules, and 0.2 to the adapter. During the online adaptation stage, the adapter was optimized with a learning rate of 1×10^{-4} , and partial or full updates were automatically triggered according to the dominant-frequency-based schedule, providing a balanced trade-off between short-term adaptability and long-term stability under concept drift.

In addition, we include nine time series models as baselines to validate the effectiveness of ADrift-B, and two adaption methods to validate the effectiveness of ADrift. All baselines were implemented with PyTorch (Autoformer [30],⁷ Informer [31],⁸ Reformer [32],⁹ Pyraformer [33],¹⁰ FEDformer [34],¹¹ PatchTST [35],¹² iTransformer

⁵ Instead of using traditional cross-validation, we adopt a sliding window approach to construct training/validation/test samples. This approach enables the model to be exposed to a wide range of temporal patterns during training, which effectively serves a similar purpose to cross-validation by enhancing generalization.

⁶ <https://open-meteo.com/>

⁷ <https://github.com/thuml/Autoformer>

⁸ <https://github.com/zhoulhuyi/Informer2020>

⁹ <https://github.com/lucidrains/reformer-pytorch>

¹⁰ <https://github.com/alipay/Pyraformer>

¹¹ <https://github.com/MAZiqing/FEDformer>

¹² <https://github.com/yuqinie98/PatchTST>

Table 2

PV power forecasting results on the overall test sets of three datasets. **Bold** values denote the best performance achieved by our backbone model (ADrift-B), and underlined values indicate the second best results. 'Imp.' reports the percentage improvement over the second best value.

Model name	Uluru		PVOD		Alice Springs	
	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓
LSTM	2.014	1.251	1.922	1.020	0.451	0.231
CNN-LSTM	2.308	1.273	1.861	0.990	0.431	0.228
Autoformer	2.778	2.056	2.053	1.131	0.474	0.269
Informer	2.751	1.530	1.952	1.081	0.442	0.227
Reformer	3.368	2.088	2.049	1.163	0.470	0.249
Pyraformer	2.801	1.593	1.817	1.022	0.459	0.224
FEDformer	2.983	2.094	1.703	0.993	0.421	0.230
PatchTST	2.117	1.243	2.185	1.106	0.497	0.256
iTransformer	2.004	<u>1.144</u>	1.879	<u>0.935</u>	0.480	0.238
DLinear	1.982	1.165	1.893	0.948	0.479	0.227
DUET	2.012	1.255	1.711	0.936	<u>0.418</u>	0.226
TFPS	1.948	1.147	1.628	0.938	0.425	<u>0.219</u>
ADrift-B	1.746	1.012	1.584	0.881	0.394	0.212
Imp.	10.37%	11.54%	2.70%	5.78%	5.74%	3.20%

[36],¹³ DLinear [37],¹⁴ DUET [17],¹⁵ TFPS [16],¹⁶ OneNet [20],¹⁷ SOLID [21],¹⁸ PROCEED [25],¹⁹ and TAFAS [24].²⁰

5.2. Performance comparison

In this paper, each model is evaluated by the mean absolute error (MAE) and root mean square error (RMSE). Each experiment is repeated five times with different random seeds for each model on our preprocessed dataset, and the mean test results are reported.

$$MAE = \frac{1}{M} \sum_{i=1}^M |\mathcal{Y}_i - \hat{\mathcal{Y}}_i|, \quad (20)$$

$$RMSE = \sqrt{\frac{1}{M} \sum_{i=1}^M (\mathcal{Y}_i - \hat{\mathcal{Y}}_i)^2}, \quad (21)$$

5.2.1. Static performance comparison (Q1)

We first evaluate the static version of our framework, ADrift-B, without any online adaptation. The comparison includes recurrent models, convolutional recurrent hybrids, and several Transformer based architectures. Experiments are conducted on three datasets (Uluru, PVOD, and Alice Springs) using three evaluation settings: the full test set (Overall), a subset of ten clear sky days (Clear Sky), and a subset of ten cloudy sky days (Cloudy Sky). These three subsets respectively represent mixed weather conditions, stable low drift conditions, and highly volatile conditions dominated by strong intra day drift.

Overall performance. As shown in Table 2, ADrift-B achieves the lowest RMSE and MAE across all three datasets even after including stronger recent baselines such as PatchTST, iTransformer, and DLinear. On Uluru, ADrift-B reduces RMSE from 1.948 to 1.746 and MAE from 1.147 to 1.012, corresponding to improvements of 10.37% and 11.54%, respectively, over the strongest competing baseline. On PVOD, the RMSE decreases from 1.628 to 1.584, yielding a 2.70% improvement, while the MAE decreases from 0.935 to 0.881, corresponding to a 5.78% improvement. On Alice Springs, ADrift-B improves RMSE from

Table 3

PV power forecasting results on the clear-sky sets of three datasets. **Bold** values denote the best performance achieved by our backbone model (ADrift-B), and underlined values indicate the second best results. 'Imp.' reports the percentage improvement over the second best value.

Model name	Uluru		PVOD		Alice Springs	
	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓
LSTM	1.651	0.996	1.824	0.883	0.387	0.164
CNN-LSTM	1.599	0.985	1.728	0.894	0.335	0.153
Autoformer	1.613	1.798	1.772	0.990	0.343	0.176
Informer	1.777	1.437	1.705	0.968	0.405	0.181
Reformer	1.966	1.571	1.818	0.984	0.387	0.149
Pyraformer	1.604	1.306	1.636	0.944	0.287	0.138
FEDformer	1.572	1.515	1.598	0.935	0.242	0.141
PatchTST	1.544	1.038	1.853	0.927	0.274	0.137
iTransformer	1.493	1.074	1.624	0.890	0.279	0.144
DLinear	<u>1.485</u>	<u>0.962</u>	1.607	0.881	0.262	0.139
DUET	1.553	0.989	<u>1.531</u>	0.889	<u>0.238</u>	<u>0.136</u>
TFPS	1.486	0.964	1.547	<u>0.872</u>	0.241	0.143
ADrift-B	1.419	0.902	1.486	0.831	0.226	0.129
Imp.	4.44%	6.24%	2.94%	4.70%	5.04%	5.15%

0.418 to 0.394 and MAE from 0.219 to 0.212, corresponding to gains of 5.74% and 3.20%, respectively. These results indicate that the proposed backbone remains consistently effective across datasets with different forecasting horizons and climatic characteristics. Notably, the superiority of ADrift-B is preserved not only against earlier Transformer-based models, but also against recently introduced strong time-series baselines, which provides stronger evidence for the effectiveness of the patch-level temporal modeling and weather-aware expert design.

Clear sky performance. Under clear-sky conditions, where PV power curves are smoother and more regular, ADrift-B still achieves the best overall performance on all three datasets, as reported in Table 3. On Uluru, the RMSE decreases from 1.485 to 1.419 and the MAE decreases from 0.962 to 0.902, corresponding to improvements of 4.44% and 6.24%, respectively. On PVOD, ADrift-B improves RMSE from 1.531 to 1.486 and MAE from 0.872 to 0.831, yielding gains of 2.94% and 4.70%, respectively. On Alice Springs, the RMSE decreases from 0.238 to 0.226, and the MAE decreases from 0.136 to 0.129, corresponding to improvements of 5.04% and 5.15%, respectively. These results suggest that even when intra-day variability is relatively weak, the proposed backbone can still learn competitive temporal representations and maintain strong predictive accuracy. Meanwhile, the fact that ADrift-B continues to outperform recently added baselines in this easier condition suggests that its gains are not limited to highly volatile scenarios, but also extend to regular clear-sky forecasting conditions.

Cloudy sky performance. As shown in Table 4, the largest performance gaps are observed under cloudy conditions, where rapid irradiance fluctuations introduce stronger intra-day non-stationarity and make forecasting substantially more difficult. Although all methods show performance degradation in this setting, ADrift-B consistently remains the best-performing model. On Uluru, ADrift-B reduces RMSE from 2.459 to 2.365 and MAE from 1.551 to 1.469, corresponding to improvements of 3.82% and 5.29%, respectively. On PVOD, the RMSE decreases from 2.546 to 2.483 and the MAE decreases from 1.562 to 1.506, yielding gains of 2.47% and 3.59%, respectively. On Alice Springs, ADrift-B improves RMSE from 1.202 to 1.146 and MAE from 1.171 to 1.142, corresponding to improvements of 4.66% and 2.48%, respectively. These results indicate that the patch-based temporal modeling together with weather-prototype-guided expert routing is particularly beneficial when the underlying input-output relationship varies rapidly within a day. These results show that cloudy conditions are substantially more challenging for all methods due to stronger intra-day fluctuations. Nevertheless, ADrift-B consistently remains the best-performing model, indicating that the proposed patch-based temporal modeling

¹³ <https://github.com/thuml/iTransformer>

¹⁴ <https://github.com/cure-lab/LTSF-Linear>

¹⁵ <https://github.com/decisionintelligence/DUET>

¹⁶ <https://github.com/syrGitHub/TFPS>

¹⁷ <https://github.com/yfzhang114/OneNet>

¹⁸ https://github.com/HALF111/calibration_CDS

¹⁹ <https://github.com/SJTU-DMTai/OnlineTSF>

²⁰ <https://github.com/kimanki/TAFAS>

Table 4

PV power forecasting results on the cloudy-sky sets of three datasets. **Bold** values denote the best performance achieved by our backbone model (ADrift-B), and underlined values indicate the second best results. 'Imp.' reports the percentage improvement over the second best value.

Model Name	Uluru		PVOD		Alice Springs	
	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓
LSTM	2.557	1.591	2.827	1.664	1.411	1.283
CNN-LSTM	4.799	3.079	2.943	1.737	1.307	1.272
Autoformer	2.810	2.116	3.124	1.598	1.775	1.652
Informer	5.142	3.936	2.741	1.791	1.204	1.176
Reformer	3.549	2.285	2.933	1.897	1.696	1.487
Pyraformer	2.972	1.732	2.578	1.881	1.593	1.428
FEDformer	3.197	2.163	2.661	1.755	1.211	1.198
PatchTST	2.895	1.833	2.653	1.596	1.213	1.187
iTransformer	2.671	1.736	2.661	1.591	1.221	1.195
Dlinear	2.536	1.658	2.574	1.583	1.218	1.183
DUET	2.468	1.576	2.552	1.574	<u>1.202</u>	1.174
TFPS	2.459	<u>1.551</u>	2.546	1.562	1.209	<u>1.171</u>
ADrift-B	2.365	1.469	2.483	1.506	1.146	1.142
Imp.	3.82%	5.29%	2.47%	3.59%	4.66%	2.48%

Table 5

Overall forecasting results on the three PV datasets, comparing (i) the static backbone (ADrift-B), (ii) the proposed adaptive framework (ADrift), and (iii) adaptive baselines formed by combining TFPS with four online adaptation baselines, including OneNet, SOLID, PROCEED and TAFAS. Both RMSE and MAE are reported. **Bold** values denote the best results, underlined values indicate the second-best results, and 'Imp.' reports the percentage improvement of ADrift relative to the second-best value.

Model name	Uluru		PVOD		Alice Springs	
	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓
TFPS+OneNet	1.882	1.131	1.604	0.912	0.418	0.211
TFPS+SOLID	1.875	1.129	1.596	0.905	0.413	0.201
TFPS+PROCEED	1.866	1.124	1.592	0.907	0.411	0.205
TFPS+TAFAS	<u>1.731</u>	<u>1.003</u>	<u>1.575</u>	<u>0.876</u>	<u>0.372</u>	<u>0.197</u>
ADrift-B	1.746	1.012	1.584	0.881	0.394	0.212
ADrift	1.653	0.957	1.499	0.845	0.361	0.186
Imp.	4.51%	4.59%	4.83%	3.54%	2.96%	5.91%

and weather-prototype-guided expert routing help maintain stronger robustness under more volatile weather conditions.

In summary, ADrift-B consistently outperforms all baselines across overall, clear-sky, and cloudy-sky evaluations on the three PV datasets. The improvements in both RMSE and MAE verify that the proposed backbone provides stable and effective temporal representations under different levels of intra-day variability. After incorporating stronger recent baselines, this conclusion still holds, which further strengthens the empirical support for the proposed design. These findings also provide a solid basis for the subsequent evaluation of the full ADrift framework, where online adaptation is introduced to further address evolving distribution shifts during deployment.

5.2.2. Adaptive performance comparison (Q2)

To further evaluate the effectiveness of our adaptation mechanism, we compare the complete ADrift framework against its static counterpart (ADrift-B) and several strong adaptive baselines, including TFPS equipped with PROCEED, TAFAS, OneNet, or SOLID. In addition to the overall results on the full test sets, we further report separate evaluations on the clear-sky and cloudy-sky subsets. The results on the full test sets are summarized in Table 5, while the clear-sky and cloudy-sky results are reported in Tables 6 and 7, respectively.

ADrift vs. ADrift-B. We first evaluate the contribution of the Learnable Adapter by comparing the full model (ADrift) with its static counterpart (ADrift-B). Across all three datasets, ADrift consistently improves upon ADrift-B in both RMSE and MAE. On Uluru, RMSE decreases from

Table 6

Forecasting results on the clear-sky subsets of the three PV datasets, comparing (i) the static backbone (ADrift-B), (ii) the proposed adaptive framework (ADrift), and (iii) adaptive baselines formed by combining TFPS with four online adaptation baselines, including OneNet, SOLID, PROCEED and TAFAS. Both RMSE and MAE are reported. **Bold** values denote the best results, underlined values indicate the second-best results, and 'Imp.' reports the percentage improvement of ADrift relative to the second-best value.

Model name	Uluru		PVOD		Alice Springs	
	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓
TFPS+OneNet	1.431	0.927	1.501	0.839	0.232	0.130
TFPS+SOLID	1.425	0.919	1.493	0.826	0.224	0.128
TFPS+PROCEED	1.412	<u>0.892</u>	1.481	0.818	0.220	0.116
TFPS+TAFAS	<u>1.407</u>	0.894	<u>1.478</u>	0.815	<u>0.215</u>	<u>0.113</u>
ADrift-B	1.419	0.902	1.486	0.831	0.226	0.129
ADrift	1.391	0.884	1.472	0.815	0.213	0.112
Imp.	1.14%	0.90%	0.41%	0.37%	0.93%	0.88%

Table 7

Forecasting results on the cloudy-sky subsets of the three PV datasets, comparing (i) the static backbone (ADrift-B), (ii) the proposed adaptive framework (ADrift), and (iii) adaptive baselines formed by combining TFPS with four online adaptation baselines, including OneNet, SOLID, PROCEED and TAFAS. Both RMSE and MAE are reported. **Bold** values denote the best results, underlined values indicate the second-best results, and 'Imp.' reports the percentage improvement of ADrift relative to the second-best value.

Model name	Uluru		PVOD		Alice Springs	
	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓
TFPS+OneNet	2.289	1.385	2.418	1.471	1.091	1.038
TFPS+SOLID	2.252	1.343	2.405	1.462	1.094	1.042
TFPS+PROCEED	2.237	1.324	2.382	<u>1.445</u>	1.088	1.031
TFPS+TAFAS	<u>2.231</u>	<u>1.317</u>	<u>2.379</u>	1.453	<u>1.083</u>	<u>1.027</u>
ADrift-B	2.365	1.469	2.483	1.506	1.146	1.142
ADrift	2.154	1.282	2.326	1.391	1.024	0.983
Imp.	3.45%	2.66%	2.23%	3.74%	5.45%	4.28%

1.746 to 1.653 (5.32%) and MAE from 1.012 to 0.957 (5.43%). On PVOD, RMSE improves from 1.584 to 1.499 (5.37%) and MAE from 0.881 to 0.845 (4.09%). On Alice Springs, the improvements are more pronounced, with RMSE reduced from 0.394 to 0.361 (8.38%) and MAE from 0.212 to 0.186 (12.26%). These results indicate that the drift-aware adapter enables ADrift to adjust its parameters in response to the distributional changes that occur during deployment. We further observe that this advantage remains consistent on both weather-specific subsets. On the clear-sky subset, ADrift still improves over ADrift-B across all three datasets, although the margins are relatively modest. For example, on Uluru, RMSE decreases from 1.419 to 1.391 and MAE from 0.902 to 0.884, while on Alice Springs, RMSE decreases from 0.226 to 0.213 and MAE from 0.129 to 0.112. On the cloudy-sky subset, the gains become more pronounced: on Uluru, RMSE decreases from 2.365 to 2.154 and MAE from 1.469 to 1.282; on PVOD, RMSE decreases from 2.483 to 2.326 and MAE from 1.506 to 1.391; and on Alice Springs, RMSE decreases from 1.146 to 1.024 and MAE from 1.142 to 0.983. This trend suggests that the proposed adaptation mechanism is particularly beneficial under more challenging conditions with stronger short-term variability and distribution shifts.

ADrift vs. Baselines. Finally, we compare the full ADrift framework with state-of-the-art adaptive baselines. On the overall test sets, ADrift achieves the best RMSE on all datasets, obtaining 1.653 on Uluru (vs. 1.731 for TFPS+TAFAS), 1.499 on PVOD (vs. 1.575), and 0.361 on Alice Springs (vs. 0.372). On the clear-sky subset, ADrift remains the best-performing method overall, achieving RMSE/MAE of 1.391/0.884 on Uluru, 1.472/0.815 on PVOD, and 0.213/0.112 on Alice Springs. On the cloudy-sky subset, ADrift also consistently outperforms all competing adaptive baselines, achieving 2.154/1.282 on Uluru, 2.326/1.391

Table 8

Ablation study of the proposed ADrift framework. The table reports RMSE and MAE for the full model and four ablated variants: V1 (w/o WPE), V2 (w/o LA), V3 (Static Base), and V4 (w/o PU). Higher errors indicate degraded performance relative to the full ADrift model.

Variant	Uluru		PVOD		AliceSprings	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
ADrift	1.653	0.957	1.499	0.845	0.361	0.186
w/o WPE	2.161	1.262	1.892	1.005	0.441	0.230
w/o LA	1.981	1.199	1.775	0.979	0.438	0.225
Static Base	3.085	1.841	2.063	1.113	0.524	0.346
w/o PU	1.746	1.012	1.584	0.881	0.394	0.212

on PVOD, and 1.024/0.983 on Alice Springs. Compared with the clear-sky subset, the relative gains under cloudy conditions are generally more pronounced, which further supports our motivation that specialized temporal adaptation is more useful when weather-driven drift becomes stronger. These differences reflect the distinct adaptation signals used by the respective methods. While baselines often focus on generalized drift estimation or alternative online adjustment strategies, ADrift bases its updates on drift estimates computed directly between consecutive batches via the temporal gap attention mechanism. This allows ADrift to make parameter-space adjustments that are specifically aligned with the instance-level external concept drift observed during deployment.

5.3. Ablation study (Q3)

To assess the contribution of each component in our framework and to examine the two-fold concept drift hypothesis introduced earlier, we conduct an ablation study comparing the full model (ADrift) with four variants: **w/o WPE** (V1, removing the weather-pattern experts), **w/o LA** (V2, removing the learnable adapter), **Static Base** (V3, removing both modules), and **w/o PU** (V4, disabling the partial update mechanism). The results are summarized in Table 8.

5.3.1. Necessity of addressing concept drift (V3 vs. ADrift)

We first examine whether a static model is sufficient for PV power forecasting. The **Static Base** variant (V3), which excludes both the expert module and the adaptive update mechanism, exhibits substantially higher errors across all datasets. On Uluru, RMSE increases from 1.653 to 3.085, corresponding to an 86.6% increase. On PVOD, the RMSE rises by 37.7% (from 1.499 to 2.063), and on Alice Springs, the increase reaches 45.2% (from 0.361 to 0.524). These results indicate that static forecasting is insufficient in settings where the mapping between inputs and PV output evolves over time due to meteorological variability.

5.3.2. Validation of the two-fold drift hypothesis (V1 & V2 vs. ADrift)

We then assess whether internal and external drifts contribute independently to performance degradation:

- **Internal Drift.** Removing the weather-pattern experts (V1) leads to a notable decline in performance, even though online adaptation remains available. RMSE increases by 30.7% on Uluru (from 1.653 to 2.161) and by 26.2% on PVOD (from 1.499 to 1.892). This suggests that modeling intra-day weather patterns is essential for capturing short-term fluctuations, and that adaptation alone cannot compensate for the absence of a structured backbone representation.
- **External Drift.** Conversely, removing the learnable adapter (V2), while retaining the expert backbone, also results in systematic degradation. RMSE increases by 19.8% on Uluru (to 1.981) and by 21.3% on Alice Springs (to 0.438). This indicates that, even with a strong backbone, temporal gaps and supervision delays can cause external drift that necessitates explicit online updates.

Taken together, the results for V1 and V2 support the two-fold drift hypothesis: *internal drift* and *external drift* affect forecasting accuracy in complementary ways, and addressing both yields the most substantial gains.

5.3.3. Effectiveness of proactive partial updates (V4 vs. ADrift)

Finally, we examine the effect of the partial update mechanism. The **w/o PU** variant (V4), which only performs full-horizon updates, produces higher errors than the full ADrift model. Re-enabling partial updates reduces RMSE by 5.6% on Uluru and PVOD (e.g., from 1.746 to 1.653 on Uluru) and by 9.1% on Alice Springs (from 0.394 to 0.361). These findings suggest that incorporating partially observed labels allows the model to adjust more promptly to emerging distributional changes, mitigating the impact of supervision lag during deployment.

5.4. Hyperparameter analysis (Q4)

To better understand how the core components influence model behaviour, we analyze the performance trends associated with four key hyperparameters as shown in Fig. 5. The observations across Uluru, PVOD, and Alice Springs are consistent and can be summarized as follows:

- **Prototype number K .** The examined range $K \in \{4, 5, 6, 7, 8\}$ shows a clear intermediate optimum. When K is too small, the prototype space lacks the capacity to represent diverse intra-day weather patterns. As K increases, performance improves until reaching the best results around $K = 6$. Further increases lead to mild degradation, suggesting that excessive prototypes introduce redundancy and reduce the distinctiveness of patch-prototype assignments.
- **Expert number E .** Within the tested range $E \in \{2, 3, 4, 5, 6, 7\}$, ADrift achieves the most stable and accurate forecasts when E is set to moderate values. Smaller values limit the ability of experts to specialize, while larger values dilute expert training due to reduced sample allocation per expert. The resulting trend indicates that too few experts restrict model expressiveness and too many reduce specialization strength, with optimal performance appearing around $E = 4$ to 6.
- **Adapter dimension d_r .** The sweep over $d_r \in \{16, 24, 32, 48, 64\}$ reveals that moderate compression levels provide the best performance. Very small dimensions restrict the adapter's ability to encode informative drift signals. Increasing d_r initially improves performance but overly large values reduce the regularizing effect of the bottleneck and introduce noise into the modulation process. The strongest results consistently appear around $d_r = 24$ to 32, with slight dataset-specific variations.
- **Bottleneck dimension r .** Across the range $r \in \{8, 12, 16, 24, 32\}$, ADrift exhibits a stable pattern where mid-range dimensions offer the most favorable trade-off between representational flexibility and regularization. Smaller dimensions restrict the reparameterization space too severely, while very large ratios weaken the intended compression behavior of the adapter. Performance trends across all datasets show that intermediate settings such as $r = 12$ or $r = 16$ are preferable, with larger values providing no further benefit.

5.5. Weather prototype and expert behavior analysis

We visualize the pairwise MMD values between learned weather prototypes in Fig. 6. Each prototype is regarded as representing a latent weather distribution underlying PV power generation, rather than an explicitly defined cluster. The diagonal elements correspond to the self-discrepancy of each prototype and remain close to zero, indicating that samples associated with the same prototype exhibit strong internal consistency. In contrast, the off-diagonal elements reflect the distributional discrepancies between different prototypes. The consistently larger values observed across most prototype pairs suggest that the

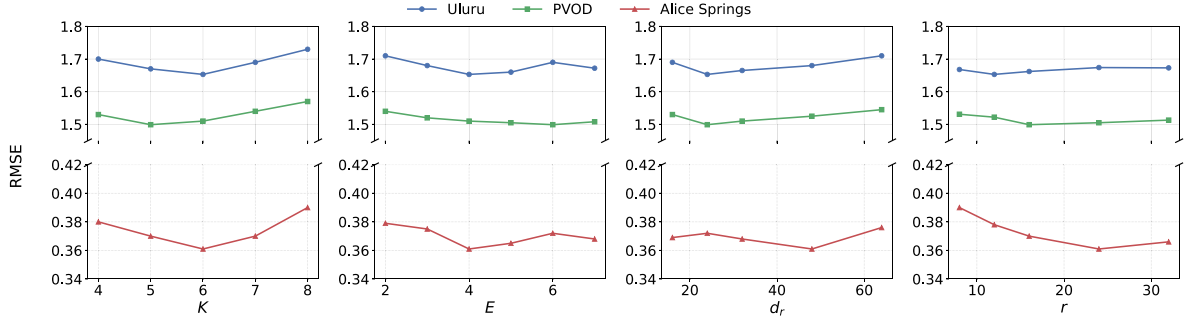


Fig. 5. Impacts of four key hyperparameters (the number of weather prototypes (K), the number of weather experts (E), and two small dimensions (d_r and r) on the performance of ADrift.

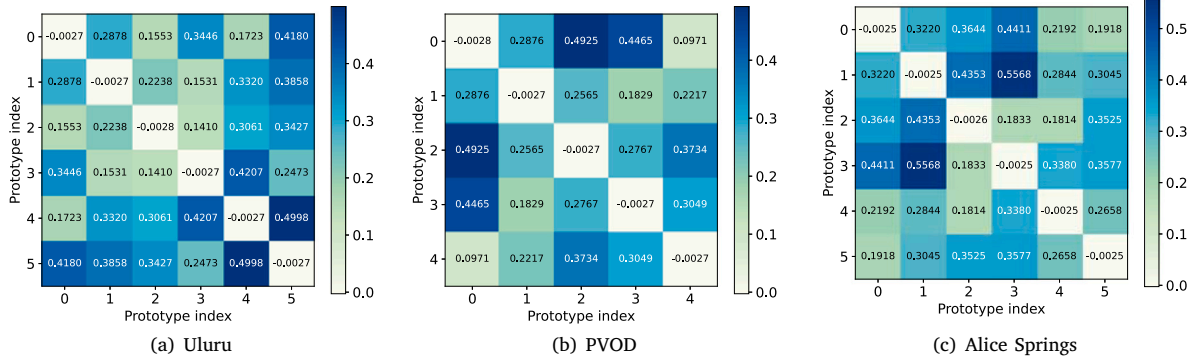


Fig. 6. Heatmap showing the MMD values of weather prototypes among three datasets.

learned prototypes capture statistically distinct weather patterns. This separation indicates that the model is able to disentangle diverse latent regimes that give rise to different PV power behaviors. From the perspective of internal concept drift, these results provide supporting evidence that the learnable weather prototypes characterize variations in the underlying weather–power relationship within the data.

Overall expert utilization. While the prototype discrepancy analysis shows that the learned prototypes are distributionally distinct, it does not directly reveal how the weather-pattern experts are utilized during routing. To further understand the expert utilization behavior of the proposed MoE design, we further report the patch-level overall top-2 selection frequency of each expert on the training set, as shown in Fig. 7. This statistic is obtained by aggregating all top-2 routing decisions over routed patches, and therefore reflects the overall expert activation distribution during optimization. The results show that, although the routing frequencies are not perfectly uniform, all experts remain meaningfully utilized, with no expert exhibiting near-zero participation.

Prototype-conditioned expert behavior. Beyond aggregate utilization, we further analyze whether different experts are preferentially activated under different prototypes. Specifically, on the test set, each patch is assigned to its dominant prototype, and the two experts selected by the router are recorded. Based on these routing records, we compute the prototype-conditioned top-2 expert selection frequencies shown in Fig. 8. The results show that expert selection patterns vary substantially across prototypes, indicating that the experts do not behave as interchangeable replicas. For example, on Alice Springs, prototypes P2–P5 more frequently activate E2 and E4, while P6 shows a markedly stronger preference for E3. On PVOD, the routing patterns are more diverse, with P3 favoring E3, P5 favoring E4, and P4 more frequently selecting E5 and E6. On Uluru, P1–P4 tend to activate

E1 more often, whereas P5–P6 exhibit relatively stronger preference for E2. These observations suggest that the routing mechanism learns prototype-conditioned expert specialization, rather than relying on a single globally dominant expert.

5.6. Robustness of the FFT-triggered update schedule

Since the proposed partial-update mechanism relies on an FFT-triggered schedule, an important question is whether this heuristic remains effective when the PV sequence exhibits weak periodicity or irregular fluctuations. To further examine this issue, we compare the proposed FFT-triggered strategy with several fixed-interval update schedules on the PVOD dataset, which contains more diverse weather conditions and stronger seasonal variability than the other two datasets. Specifically, we consider fixed update intervals of 4, 8, 12, and 24 steps, corresponding to 1, 2, 3, and 6 h, respectively. Table 9 reports both the overall results on PVOD and a separate evaluation on cloudy days, which represent a more challenging subset with stronger short-term variability.

As shown in Table 9, the proposed FFT-triggered strategy achieves the best overall performance among all compared schedules, with an RMSE/MAE of 1.499/0.845. Compared with the strongest fixed-interval baselines, namely 12-step and 24-step updates, this result suggests that adapting the update interval to the observed temporal variation is more effective than using a manually specified schedule. On cloudy days, where irradiance variations are usually more irregular and noisy, the FFT-triggered strategy remains highly competitive. Although the 24-step baseline achieves a slightly lower RMSE (2.314 vs. 2.326), the proposed method obtains the best MAE (1.391). This indicates that, even when the dominant spectral component becomes less pronounced, the FFT-based trigger can still provide useful coarse-grained information about the variation timescale, thereby supporting effective update scheduling in practice.

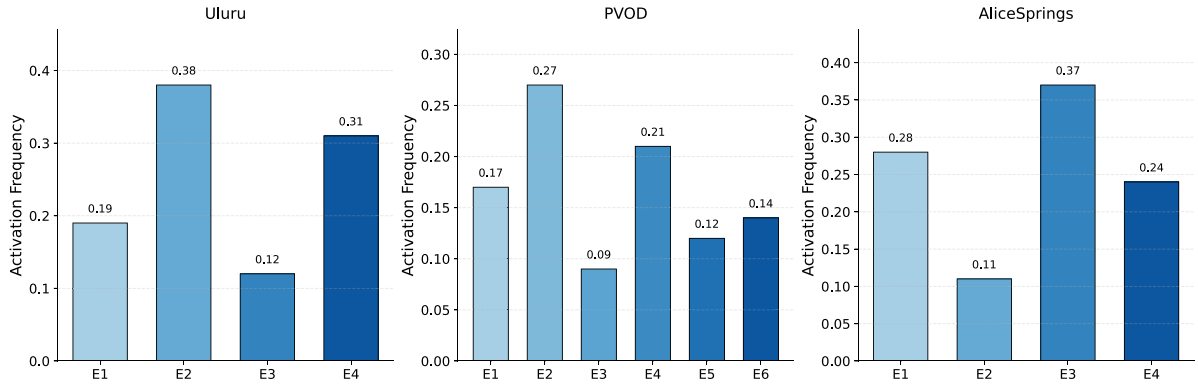


Fig. 7. Patch-level overall top-2 selection frequency of each expert on the training set. The statistic is obtained by aggregating all top-2 routing decisions over routed patches.

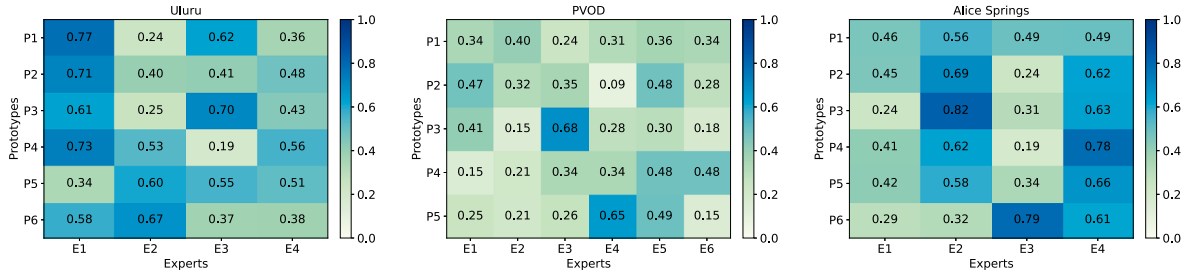


Fig. 8. Prototype-conditioned top-2 expert selection frequencies on the test set across three datasets: Alice Springs, PVOD, and Uluru. Each patch is assigned to its dominant prototype, and the two experts selected by the router are counted. The resulting heatmaps highlight how expert selection patterns vary across prototypes, suggesting conditional expert specialization.

Table 9

Comparison between fixed-interval updates and the proposed FFT-triggered update strategy on the PVOD dataset. Fixed intervals of 4, 8, 12, and 24 steps correspond to 1, 2, 3, and 6 h, respectively.

Update strategy	Overall		Cloudy	
	RMSE ↓	MAE ↓	RMSE ↓	MAE ↓
4 steps	1.516	0.903	2.346	1.439
8 steps	1.521	0.917	2.351	1.432
12 steps	<u>1.504</u>	0.859	2.337	1.412
24 steps	1.507	<u>0.852</u>	2.314	<u>1.399</u>
FFT-triggered	1.499	0.845	<u>2.326</u>	1.391

Table 10

Average per-step latency and peak memory usage on the Uluru dataset during inference-time adaptation under three execution modes: prediction-only, partial-update, and full-update.

Metric	Prediction-only	Partial-update	Full-update
Latency (ms)	19.81	23.64	24.12
Memory (GB)	0.53	0.53	0.53

moderate in practice, while memory consumption stays stable across different execution modes.

5.7. Inference-time efficiency analysis

Since ADrift performs adapter updates during inference, it is necessary to quantify the associated computational overhead. To this end, we report the *average per-step latency* (ms) and *peak memory usage* (GB) during the inference-adaptation phase on the Uluru dataset. The reported latency is averaged over all inference steps under each execution mode.

At inference time, ADrift may operate in three different execution modes depending on whether delayed supervision is available. Specifically, *Prediction-only* denotes the case where the model directly produces forecasts without updating the adapter; *Partial-update* denotes the case where the adapter is updated once partially observed ground truth becomes available; and *Full-update* denotes the case where the adapter is updated after the full ground truth of the current step becomes available.

The results are summarized in Table 10. The average per-step latency is 19.81 ms for Prediction-only, 23.64 ms for Partial-update, and 24.12 ms for Full-update. Meanwhile, the peak memory usage remains constant at 0.53 GB across all three modes. These results indicate that inference-time adaptation introduces additional computation compared with direct prediction, but the extra overhead remains

5.8. Case study of weather transition

We further examine a challenging inter-day transition scenario on the PVOD dataset, where the previous day is relatively smooth but the current day exhibits substantially stronger fluctuations. Such a case creates a clear mismatch between the lagged reference and the current weather pattern, and is therefore useful for assessing whether the fixed-lag design of TGA leads to cold-start behavior in the early stage of a new day. As shown in Fig. 9, although both models become less accurate at the beginning of the transition day, ADrift does not degenerate into a purely cold-start regime. The offline backbone and weather-pattern experts still provide a meaningful prior, and once delayed observations become available, the online adapter updates enable ADrift to recover faster than ADrift-B and to follow the rapidly varying PV trajectory more closely.

6. Discussion

The proposed ADrift framework integrates weather-aware representation learning with drift-guided online adaptation. While the method demonstrates strong performance across diverse sky conditions and

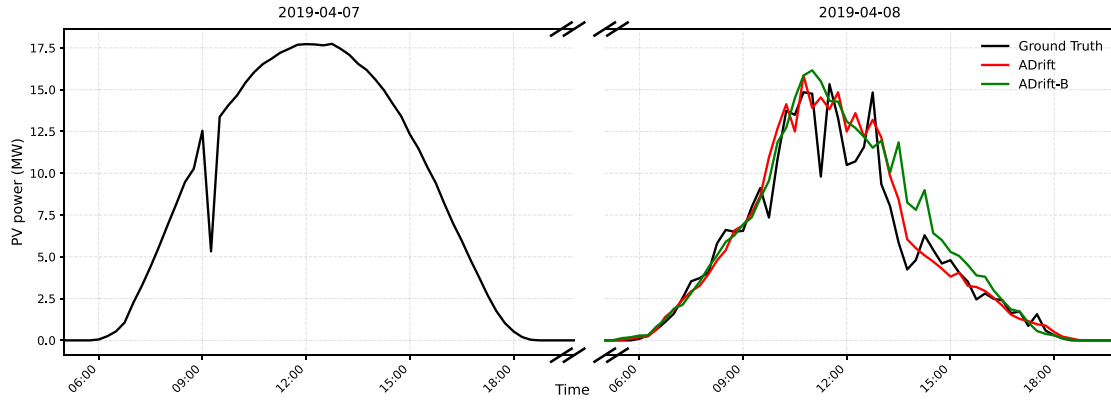


Fig. 9. Case study on two consecutive days from the PVOD dataset under an abrupt inter-day weather transition. The first day shows the lagged-reference PV profile, while the second day presents the ground truth together with the predictions of ADrift and ADrift-B. The example is used to examine whether a mismatched lagged reference causes cold-start behavior or a pronounced error increase during the early stage of a new weather pattern.

effectively mitigates both internal and external concept drift, several advantages and limitations merit discussion.

Empirical evidence for learning the two types of drift. Beyond the overall forecasting improvements, the experimental results also provide more explicit evidence that ADrift addresses the two types of concept drift considered in this work. For *internal drift*, defined as intra-day changes in PV generation patterns under evolving weather conditions, the comparison between **ADrift-B** and **w/o WPE** shows that removing the Weather-Pattern Experts consistently degrades performance, indicating that modeling multiple weather-dependent mappings is beneficial. In addition, the **prototype-conditioned top-2 expert selection frequencies** across the three datasets show that different prototypes exhibit different expert selection preferences, suggesting that the routing mechanism learns meaningful expert specialization rather than collapsing to a single static behavior. For *external drift*, defined here as the deployment-oriented mismatch caused by train-test temporal gaps and delayed supervision, several results support the effectiveness of the online adaptation design. The comparison between **ADrift** and **ADrift-B** shows that introducing the learnable adapter improves forecasting accuracy over the static backbone, while the **w/o PU** ablation further highlights the importance of timely partial updates. Moreover, the comparison between **fixed-interval** and **FFT-triggered** updating suggests that the proposed update schedule is more effective than a naive periodic scheme. Finally, the **case study on two consecutive days from the PVOD dataset** provides intuitive evidence that ADrift can recover faster under post-deployment temporal changes and follow volatile PV trajectories more closely once new observations become available. Together, these results make the respective roles of the WPE module and the adapter-based online update mechanism more transparent in handling internal and external drift.

Strengths. A key strength of ADrift lies in its structural disentanglement of drift sources. By employing patch-based weather experts to handle internal drift and a learnable adapter to address external drift, the framework avoids the interference commonly seen when all drift types are handled by a unified adaptation module. Specifically, the prototype-guided expert routing mechanism preserves physical interpretability within the backbone, anchoring predictions to distinct meteorological patterns. Moreover, the drift-aware parameter modulation provides a more fundamental solution than simple input-space adjustment. By directly modulating backbone weights using the temporal gap attention signal, the model can adjust its underlying hypothesis function to match evolving data distributions. Finally, the proactive partial-update strategy, scheduled by frequency-domain analysis, effectively mitigates the supervision delay problem, enabling timely adaptation to transient fluctuations without requiring full-horizon ground truth.

Limitations. Despite its promising performance, ADrift has several limitations. First, the computational latency introduced by online updates is non-negligible. Although the adapter is parameter-efficient, iterative gradient-based updates during inference incur overhead that may challenge deployment on resource-constrained edge devices. Second, the effectiveness of the WPE backbone depends on the diversity of historical data. If the training set lacks representative extreme weather events, the learned prototypes may fail to cover the full spectrum of meteorological patterns, limiting generalization. Third, while effective for short-term volatility, the current drift estimation mechanism is purely temporal and does not incorporate spatial or exogenous cues (such as cloud motion vectors from satellite imagery), which could provide anticipatory signals for approaching weather transitions.

Applicability and future directions. ADrift is particularly suitable for high-resolution, high-volatility PV forecasting scenarios where cloud-driven fluctuations dominate. In highly stable climates, the benefits of such fine-grained adaptation may be marginal relative to the computational cost. Future work will explore two directions: (1) **Gradient-free and More Efficient Adaptation:** Investigating meta-learning or hypernetwork-based mechanisms to predict optimal adapter parameters directly from drift signals, thereby avoiding the cost of explicit gradient computation during inference. Such a direction may further improve the practicality of drift-aware forecasting in real-time deployment scenarios. (2) **Generalization to Multi-site and Multi-market Forecasting:** Extending the framework beyond single-task PV forecasting to more heterogeneous forecasting environments. In particular, electricity market forecasting has recently attracted increasing attention across day-ahead, intraday, and balancing settings, where data distributions can also vary substantially across time and market conditions. Recent studies [38–40], suggest that electricity price prediction involves strong temporal variability, multi-source inputs, and regime-dependent behavior. These characteristics make it a promising direction for extending drift-aware modeling from PV forecasting to broader energy forecasting tasks. A recent survey [41], also highlights the diversity and evolving challenges of such markets, motivating future investigation into whether mechanisms such as weather-pattern experts and adapter-based online updates can be generalized to market-specific latent patterns and rapidly changing trading environments.

7. Conclusion

This work presented ADrift, a unified framework for short-term photovoltaic power forecasting under dynamic and volatile weather conditions. The method integrates patch-based temporal modeling, weather prototype-guided experts, and a drift-aware adapter to jointly address internal and external concept drift. By disentangling drift sources and

introducing a temporal gap attention mechanism for drift estimation, ADrift enables lightweight and targeted online updates without incurring substantial computational overhead. Extensive experiments across multiple datasets and sky conditions indicate that ADrift can improve forecasting performance, particularly in cloudy and highly variable scenarios where distributional changes are more pronounced. The proactive partial-update strategy further helps mitigate supervision delays, allowing the model to respond more promptly to transient fluctuations.

Despite these advantages, several challenges remain, including the computational latency of online updates, dependence on sufficiently diverse historical weather patterns, and the lack of spatial cues in drift estimation. Addressing these limitations offers promising directions for future work. In particular, gradient-free adaptation mechanisms and multi-site prototype sharing may further improve the scalability and generalization of the framework. Overall, ADrift provides a principled and flexible basis for drift-aware PV forecasting and may support more reliable renewable energy integration under real-world uncertainty.

CRedit authorship contribution statement

Wentao Wang: Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Haiyan Lu:** Writing – review & editing, Supervision, Conceptualization. **Ayesha Ubaid:** Writing – review & editing, Resources. **Fanyi Yang:** Writing – review & editing. **Runyao Yu:** Writing – review & editing. **Jianzhou Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] E. Lopez-Ortiz, M. Jiménez, L.M. Soria-Morillo, J.A. Alvarez-Garcia, J.J. Vegas-Olmos, Exploring low-resource weather forecasting with echo state network-based architectures and satellite data, *Knowl.-Based Syst.* (2025) 113692.
- [2] Yuhao Nie, Ahmed S. Zamzam, Adam Brandt, Resampling and data augmentation for short-term PV output prediction based on an imbalanced sky images dataset using convolutional neural networks, *Sol. Energy* 224 (2021) 341–354.
- [3] Guomin Xie, Zijian Zhang, Sen Xie, Zhaowei Yuan, Hao Liu, CPWformer-DEC: improved transformer with class-priority weather attention and dynamic error compensation for photovoltaic power forecasting, *Expert Syst. Appl.* 301 (2026) 130580.
- [4] Wentao Wang, Haiyan Lu, Ayesha Ubaid, Fanyi Yang, Yan Zhao, Jianzhou Wang, Volatility-aware sample re-weighting framework for short-term photovoltaic power forecasting, *Inf. Process. Manage.* 63 (4) (2026) 104612.
- [5] Yang Lin, Irena Koprinska, Mashud Rana, Temporal convolutional neural networks for solar power forecasting, in: 2020 International Joint Conference on Neural Networks, IJCNN, IEEE, 2020, pp. 1–8.
- [6] Ali Agga, Ahmed Abbou, Moussa Labbadi, Yassine El Houm, Short-term self consumption PV plant power production forecasts based on hybrid CNN-LSTM, ConvLSTM models, *Renew. Energy* 177 (2021) 101–112.
- [7] Rujian Chen, Gang Liu, Yisheng Cao, Gang Xiao, Jianchao Tang, CGAformer: Multi-scale feature transformer with MLP architecture for short-term photovoltaic power forecasting, *Energy* 312 (2024) 133495.
- [8] Wentao Wang, Haiyan Lu, Ayesha Ubaid, Jianzhou Wang, A complementary time-frequency domain approach for short-term photovoltaic power forecasting, *J. Supercomput.* 81 (11) (2025) 1–31.
- [9] En Yu, Jie Lu, Xiaoyu Yang, Guangquan Zhang, Zhen Fang, Learning robust spectral dynamics for temporal domain generalization, in: *Advances in Neural Information Processing Systems*, 2025.
- [10] Xing Luo, Dongxiao Zhang, An adaptive deep learning framework for day-ahead forecasting of photovoltaic power generation, *Sustain. Energy Technol. Assess.* 52 (2022) 102326.
- [11] Le Zhang, Jizhong Zhu, Di Zhang, Yun Liu, An incremental photovoltaic power prediction method considering concept drift and privacy protection, *Appl. Energy* 351 (2023) 121919.
- [12] Siyuan Mu, Sen Lin, A comprehensive survey of mixture-of-experts: Algorithms, theory, and applications, 2025, arXiv preprint arXiv:2503.07137.
- [13] Hongbo Li, Sen Lin, Lingjie Duan, Yingbin Liang, Ness B. Shroff, Theory on mixture-of-experts in continual learning, in: *The Thirteenth International Conference on Learning Representations*, 2024.
- [14] Van-Tuan Tran, Quoc-Viet Pham, et al., Revisiting sparse mixture of experts for resource-adaptive federated fine-tuning foundation models, in: *ICLR 2025 Workshop on Modularity for Collaborative, Decentralized, and Continual Deep Learning*, 2025.
- [15] En Yu, Jie Lu, Kun Wang, Xiaoyu Yang, Guangquan Zhang, Drift-aware collaborative assistance mixture of experts for heterogeneous multistream learning, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026.
- [16] Yanru Sun, Zongxia Xie, Emadeldeen Eldele, Dongyue Chen, Qinghua Hu, Min Wu, Learning pattern-specific experts for time series forecasting under patch-level distribution shift, in: *Advances in Neural Information Processing Systems*, 2025.
- [17] Xiangfei Qiu, Xingjian Wu, Yan Lin, Chenjuan Guo, Jilin Hu, Bin Yang, Duet: Dual clustering enhanced multivariate time series forecasting, in: *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 1*, 2025, pp. 1185–1196.
- [18] Steven C.H. Hoi, Doyen Sahoo, Jing Lu, Peilin Zhao, Online learning: A comprehensive survey, *Neurocomputing* 459 (2021) 249–289.
- [19] Quang Pham, Chenghao Liu, Doyen Sahoo, Steven Hoi, Learning fast and slow for online time series forecasting, in: *The Eleventh International Conference on Learning Representations*, 2023.
- [20] Qingsong Wen, Weiqi Chen, Liang Sun, Zhang Zhang, Liang Wang, Rong Jin, Tieniu Tan, et al., Onenet: Enhancing time series forecasting models under concept drift by online ensembling, *Adv. Neural Inf. Process. Syst.* 36 (2023) 69949–69980.
- [21] Mouxiang Chen, Lefei Shen, Han Fu, Zhuo Li, Jianling Sun, Chenghao Liu, Calibration of time-series forecasting: Detecting and adapting context-driven distribution shift, in: *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2024, pp. 341–352.
- [22] Daojun Liang, Haixia Zhang, Jing Wang, Dongfeng Yuan, Minggao Zhang, Act now: A novel online forecasting framework for large-scale streaming data, 2024, arXiv preprint arXiv:2412.00108.
- [23] Ying-yei Ava Lau, Zhiwen Shao, Dit-Yan Yeung, Fast and slow streams for online time series forecasting without information leakage, in: *The Thirteenth International Conference on Learning Representations*, 2025.
- [24] HyunGi Kim, Siwon Kim, Jisoo Mok, Sungroh Yoon, Battling the non-stationarity in time series forecasting via test-time adaptation, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 17, 2025, pp. 17868–17876.
- [25] Lifan Zhao, Yanyan Shen, Proactive model adaptation against concept drift for online time series forecasting, in: *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 1*, 2025, pp. 2020–2031.
- [26] Xinyu Wang, Wenping Ma, A hybrid deep learning model with an optimal strategy based on improved VMD and transformer for short-term photovoltaic power forecasting, *Energy* 295 (2024) 131071.
- [27] Pengyun Jia, Haibo Zhang, Xinmiao Liu, Xianfu Gong, Short-term photovoltaic power forecasting based on VMD and ISSA-GRU, *IEEE Access* 9 (2021) 105939–105950.
- [28] Tiechui Yao, Jue Wang, Haoyan Wu, Pei Zhang, Shigang Li, Yangang Wang, Xuebin Chi, Min Shi, A photovoltaic power output dataset: Multi-source photovoltaic power output dataset with python toolkit, *Sol. Energy* 230 (2021) 122–130.
- [29] Jing Yang, Hui He, Xueming Zhao, Jue Wang, Tiechui Yao, Haizhou Cao, Meng Wan, Day-ahead PV power forecasting model based on fine-grained temporal attention and cloud-coverage spatial attention, *IEEE Trans. Sustain. Energy* (2023).
- [30] Haixu Wu, Jiehui Xu, Jianmin Wang, Mingsheng Long, Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting, *Adv. Neural Inf. Process. Syst.* 34 (2021) 22419–22430.
- [31] Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, Wancai Zhang, Informer: Beyond efficient transformer for long sequence time-series forecasting, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, 2021, pp. 11106–11115.
- [32] Nikita Kitaev, Łukasz Kaiser, Anselm Levskaya, Reformer: The efficient transformer, 2020, arXiv preprint arXiv:2001.04451.
- [33] Shizhan Liu, Hang Yu, Cong Liao, Jianguo Li, Weiya Lin, Alex X. Liu, Schahram Dustdar, Pyraformer: Low-complexity pyramidal attention for long-range time series modeling and forecasting, in: *International Conference on Learning Representations*, 2022.
- [34] Tian Zhou, Ziqing Ma, Qingsong Wen, Xue Wang, Liang Sun, Rong Jin, Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting, in: *International Conference on Machine Learning*, PMLR, 2022, pp. 27268–27286.
- [35] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, Jayant Kalagnanam, A time series is worth 64 words: Long-term forecasting with transformers, in: *International Conference on Learning Representations*, 2023.

- [36] Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, Mingsheng Long, Itransformer: Inverted transformers are effective for time series forecasting, in: *The Twelfth International Conference on Learning Representations*, 2024.
- [37] Ailing Zeng, Muxi Chen, Lei Zhang, Qiang Xu, Are transformers effective for time series forecasting? in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 9, 2023, pp. 11121–11128.
- [38] Runyao Yu, Chenhui Gu, Jochen Stiasny, Qingsong Wen, Wasim Sarwar Dilov, Lianlian Qi, Jochen L. Cremer, Pricefm: Foundation model for probabilistic electricity price forecasting, 2025, arXiv preprint [arXiv:2508.04875](https://arxiv.org/abs/2508.04875).
- [39] Runyao Yu, Yuchen Tao, Fabian Leimgruber, Tara Esterl, Jochen Stiasny, Derek W. Bunn, Qingsong Wen, Hongye Guo, Jochen L. Cremer, OrderFusion: Encoding orderbook for end-to-end probabilistic intraday electricity price forecasting, 2025, arXiv preprint [arXiv:2502.06830](https://arxiv.org/abs/2502.06830).
- [40] Runyao Yu, Ruochen Wu, Yongsheng Han, Jochen L. Cremer, Orderbook feature learning and asymmetric generalization in intraday electricity markets, 2025, arXiv preprint [arXiv:2510.12685](https://arxiv.org/abs/2510.12685).
- [41] Runyao Yu, Derek W. Bunn, Julia Lin, Jochen Stiasny, Fabian Leimgruber, Tara Esterl, Yuchen Tao, Lianlian Qi, Yujie Chen, Wentao Wang, et al., Deep learning for electricity price forecasting: A review of day-ahead, intraday, and balancing electricity markets, 2026, arXiv preprint [arXiv:2602.10071](https://arxiv.org/abs/2602.10071).